

Building Success: The Effect of On-Campus Housing Expansions on Student Outcomes

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Abstract

Housing shortages are increasingly cited as barriers to degree completion. Using a 17-year panel of 1,256 U.S. colleges, I estimate the effect of dormitory expansions on retention and graduation via within-college differencing models, confirmed by stacked difference-in-differences. Large expansions do not affect first-year retention but raise four-year graduation by 1–2.25 percentage points; effects grow with expansion size and cohort exposure and reflect higher total completion rather than re-timing. Mechanism analyses find no rent-market spillovers, pointing to campus housing access as the primary channel. Expanding on-campus capacity is a tractable, institution-controlled policy to raise graduation rates.

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1. Introduction

Rising tuition and living costs present one of the most significant barriers to higher education attainment in the United States and globally. High housing costs are widely cited as obstacles to effective learning, due to long commutes, the need to work more hours, and reduced attendance.² As rental markets tighten, these financial and logistical pressures on persistence and completion intensify. Colleges have a policy lever at their disposal: the expansion of on-campus housing capacity. When institutions build new residence halls or substantially enlarge existing ones, they not only offer more students the convenience and support of campus living but also relieve pressure on the local rental market. The dual impact of such an expansion may translate into improved retention and graduation outcomes, directly through greater campus residency and indirectly through lower off-campus rents.

To my knowledge, this is the first paper to estimate the causal effect of college housing expansions on student outcomes. I investigate how large expansions in on-campus student housing affect two critical measures of student success: first-to-second-year retention and four-year graduation rates. Completion is a particularly important margin: a large literature shows that obtaining a degree is a primary determinant of long-run earnings, labor market attachment, and economic mobility (Arcidiacono 2004; Arcidiacono et al. 2010; Oreopoulos and Salvanes 2011). By treating residence hall openings and expansions as discrete shocks to housing supply, the results address a practical policy question: can colleges improve student outcomes by strategically investing in housing infrastructure?

The primary empirical strategy is within-college difference models spanning 17 years and 1,256 colleges. Average effects on first-year retention are indistinguishable from zero, while four-year graduation rises by 1-2.25 percentage points, with effects increasing in the size of the expansion. A stacked difference-in-differences applied to a restricted subset of colleges with single, clearly timed capacity jumps confirms these patterns. These results are consistent with a simple explanation: because most institutions already prioritize

²See, for example, <https://apnews.com/article/rent-us-college-students-1cc3bc02f25f8a628f828902029bad9b> in the U.S., <https://www.theguardian.com/money/article/2024/aug/18/have-laptop-can-travel-the-rise-and-rise-of-the-commuter-student> in the UK, and <https://www.irishtimes.com/ireland/education/2025/02/11/high-rental-costs-are-pricing-out-thousands-of-students-and-risk-increased-college-dropout-rates/> in Ireland.

first-year housing, additional beds primarily relax constraints for upper-year students, improving completion. Comparing cohorts that were already enrolled at the time of an expansion with a pre-expansion cohort shows that effects are larger for students with greater exposure to the additional capacity, while similar effects on four-, six-, and eight-year graduation rates suggest that expansions raise total completion rather than merely accelerate it. I also examine heterogeneity across institutional subgroups and test for spillovers to nearby colleges; these results provide little systematic evidence of effects operating through local rent markets, pointing instead to a primary role for the benefits of campus residence.

2. Conceptual Framework & Institutional Background

College housing expansions can affect student success through two distinct but potentially overlapping channels. First, the “campus residence channel”: when a new residence hall opens, more students can live on campus. This is expected to improve academic outcomes in several ways that prior work has shown to matter for student success: peer interactions, greater academic structure and monitoring, access to support services, and increased likelihood of attending class (see, for example, Angrist et al. (2009); Dobkin et al. (2010); Sacerdote (2001); Stinebrickner and Stinebrickner (2006) for related evidence on these mechanisms). On-campus residence could also harm academic performance if greater involvement in campus social life crowds out study time. Because U.S. colleges typically prioritize first-years for dorm beds,³ one would expect an effect on first-year retention rates through the campus residence channel. However, if prioritization means that most first-years are already housed, expansions may instead benefit older students at the margin: the proportion of first-years accommodated remains constant, while additional spaces become available for upper-year students.

Second, the rent market channel: by increasing on-campus capacity, col-

³For example, among the ten largest colleges in the sample (by mean enrollment), eight explicitly refer to designated freshmen housing areas, priority given to freshmen, or requirements for freshmen to live on campus on their websites (Ohio State University, University of Central Florida, Grand Canyon University, University of Minnesota – Twin Cities, University of Texas at Austin, Michigan State University, New York University, and University of Illinois at Urbana-Champaign), while only two do not (University of Florida and Texas A&M University – College Station allocate housing on a first-come, first-served basis regardless of seniority).

leges effectively remove a number of students from the external market, decreasing the demand for private rentals near the campus. All else equal, this would be expected to reduce equilibrium rents in the off-campus market, and improve the availability of housing near the campus (reducing search costs and commute times) for all students (particularly students in upper years who tend to live off campus). The reduction in rental costs may also allow these students to substitute in study for part-time work, thereby improving persistence and completion.⁴

Allocation policies vary widely: many institutions guarantee (or in some cases even require) on-campus housing for first-year students; continuing-student housing is commonly allocated through lotteries with class-year priorities and limited guarantees. Policies frequently reserve a portion of beds for specific programs or groups (such as international students), with the residual stock assigned by lottery among second-years and above.⁵ This institutional variation determines which cohorts receive the marginal beds when capacity expands and therefore which outcomes can move first. For example, expansions that primarily ease first-year constraints are likely to be reflected in first-year retention outcomes via the campus residence channel, whereas expansions that free up places for upper-year students will initially affect upper-year outcomes through a combination of the campus residence and rent market channels.

The two channels imply different timing patterns for observable effects. First-year retention effects should arise as soon as new capacity comes on-line, assuming dorms are fully operational by the start of the academic year. Graduation effects will emerge more gradually, as successive cohorts benefit from expanded capacity across multiple years of study. The impact on four-year graduation rates will therefore depend on both the timing of expansions and the institution's housing allocation policies, which determine how capacity is distributed across student years.

The U.S. undergraduate sector is not uniformly residential: among full-time undergraduates in 2015-16, roughly 32% lived on campus, 44% lived

⁴In the 2012 National Postsecondary Student Aid Study, approximately two-thirds of undergraduates report paying at least some of their own rent or mortgage while enrolled ([National Center for Education Statistics 2014](#)).

⁵When students do not receive an institutional bed, U.S. colleges typically do not subsidize off-campus rent nor provide special assistance; the outside option is the private market.

off-campus away from parents, and 24% lived with parents.⁶ Patterns differ somewhat outside the U.S. In Eurostudent countries, for example, an average 34% of students live with their parents and only about 15% reside in designated student housing (Hauschildt et al. 2024). Where a large share of students commute from the parental home or rely on the local private rental market, the rent market channel is likely to be relatively more salient than the campus residence channel. The framework is therefore applicable outside the U.S., although the relative magnitudes of effects will depend on local housing institutions.

3. Literature Review

Research on students’ residential arrangements dates back more than five decades (see Blimling (1989) for an early review), yet robust causal evidence is limited. Broadly, the empirical literature divides into three strands: (i) single-institution causal studies, (ii) multi-institution evidence using national datasets, and (iii) work on institutional spending that implicitly touches on housing issues. None of these strands, however, evaluates the consequences of expanding residence-hall capacity, the policy lever which is the focus of this study.

Several papers exploit within-college variation to estimate the effect of living on campus. Reynolds (2020) uses a natural experiment at Kent State whereby some students are exempt from a residency rule, and shows that on-campus residence raises both grade point average (GPA) and graduation rates. Similarly, Araujo and Murray (2010) instrument for students’ residence choices at a large public university and find GPA gains of 0.2–1.0 points. In a randomized trial, Perez-Felkner et al. (2022) offer rent-free apartments to community-college students. Participants work fewer paid hours and are more likely to persist to second year. These studies deliver credible estimates of the campus residence channel (the benefits that arise from proximity to peers, classes and campus support) but say nothing about the effects of how many beds a college provides or about spillovers to the local rental market.

Moving beyond single campuses, Schudde (2011) combines the Educational Longitudinal Study with propensity-score matching and finds that students who live on campus are three percentage points more likely to en-

⁶See National Center for Education Statistics Profile of Undergraduate Students: <https://nces.ed.gov/pubs2019/2019467.pdf>.

roll in their second year. Descriptive national analyses yield mixed results: positive associations are found by Bozick (2007) and Thompson et al. (1993), but no significant effects remain once covariates are added in Lopez Turley and Wodtke (2010). Although these studies cover hundreds of institutions, their treatment remains residence status, not housing supply.

Two papers that use IPEDS (Integrated Postsecondary Education Data System) data on U.S. colleges address housing indirectly. Webber and Ehrenberg (2010) relate functional expenditures to graduation and persistence, finding that spending on student services matters, but auxiliary spending (including housing) is omitted from their models, likely because auxiliary units (such as residence halls and dining) are operated as self-supporting enterprises whose expenditures are offset by their own revenues. Zong and Davis (2024) include dormitory variables in an attempt to model retention and graduation, but conclude that the effect of capacity is controversial and requires further study.

Guzman (2021) is to the best of my knowledge the only study to exploit national IPEDS panel data to address a question of this nature with causal intent. Guzman analyzes mandatory residency policies for 100 top universities and, after adding institution fixed effects, finds no impact on retention or six-year completion. The bed-to-enrollment ratio appears as a control, but dormitory construction or expansion does not serve as the treatment.

Across all of this work, treatment variation comes from whether students reside on campus, not from increases in the number of beds. The literature therefore provides evidence on the campus residence channel (although the populations differ) but is silent on the rent market channel (the idea that new dorms ease pressure on local housing markets and benefit off-campus students). By treating residence-hall openings as discrete shocks, this study estimates the combined effects of both channels and relates directly to a policy lever (expanding capacity) that colleges can control.

4. Data

4.1. Overview

I construct a panel of 1,256 U.S. four-year colleges⁷ from 2003 to 2019 (21,352 observations), drawing on three data sources. The core dataset is IPEDS ([National Center for Education Statistics 2026](#)), which covers institutions participating in federal student aid programs accounting for almost all public and private colleges. I supplement this with American Community Survey (ACS) rent data ([U.S. Census Bureau 2020](#)) to measure local housing market conditions and Census Bureau Building Permits Survey (BPS) data ([U.S. Census Bureau 2021](#)) to proxy private rental supply. The IPEDS panel is constructed by merging the following IPEDS survey components:

- IPEDS Institutional Characteristics data files: Provide information on each institution’s housing capacity, annual room charges, and mandatory residence rules.
- IPEDS Institutional Directory data files: Provide institution-level characteristics including public/private control, urban/rural locale classification, and ZIP code.
- IPEDS Admissions and Test Scores data files: Provide median and quartile ranges for SAT scores and ACT scores for incoming students.
- IPEDS Graduation Rate data files: Provide the percentage of students in each cohort in each institution to graduate within 100% of normal time (4 years), 150% of normal time (6 years) and 200% of normal time (8 years). Graduation rates for each cohort are reported 8 years after enrollment, meaning that this dataset captures cohort graduation outcomes through 2015 entrants.
- IPEDS fall enrollment data files: Provide first-to-second-year retention rates, total number of incoming first-years, and total headcount for each institution.
- IPEDS finance data files: Provide total spending on instruction, academic support, student services, and scholarships in each year.
- IPEDS Student Financial Aid files: Provide information on the percentage of students in each institution in receipt of Pell grants.

⁷Specifically, the sample is restricted to colleges which are classified as four-year in every year of the sample period. When extended to two-year colleges, the analysis is severely underpowered because relatively few two-year institutions offer on-campus housing.

I restrict the sample to relatively large institutions (average enrollment $\geq 1,000$, and further dropping another 50 colleges which report an incoming cohort of less than 50 students in some year over the sample period). Institutions that never offer housing during the sample period are dropped. I exclude large institutions which mainly cater to students online (Arizona State Digital Immersion, Southern New Hampshire University, and Liberty University). I drop 2020 onward to avoid COVID-19 confounding.

I conduct basic integrity checks across IPEDS files (e.g. internal consistency of year-to-year capacity, retention and graduation). On this basis, one anomalous capacity report was corrected after manual inspection.⁸ I further drop 54 colleges where any entering cohort in the sample period has a reported retention or graduation rate of 0%, for two reasons: (i) concerns over misreporting, and (ii) a lack of within-institution variation in the outcomes of interest over time. Results are robust to these sampling restrictions. Classical mismeasurement in the treatment (capacity) would attenuate estimates toward zero, so any positive effects despite potential noise likely understate true impacts. In addition, reported decreases may sometimes reflect ownership or accounting changes (e.g., transfers to affiliated entities) rather than physical closures, which would also bias estimates toward zero. IPEDS non-response arises primarily from administrative reporting gaps (such as institutions missing a filing deadline) rather than underlying student outcomes, so any missingness is plausibly missing at random. Following these sampling restrictions, the sample captures the vast majority of U.S. 4-year institutions. For instance, in 2019, it represented 71% of the total undergraduate enrollment at 4-year institutions as reported by IPEDS.

4.2. Descriptive Statistics

Table 1 shows summary statistics for the variables used throughout the analysis.

The number of beds offered per college ranges from 0 (as although colleges are dropped from the sample if they never offer housing, some do not offer housing in every sample year) to 19,446 (the capacity in Michigan State University in 2004). Housing capacity refers to institutionally-controlled residential facilities; institutions report whether they provide institutionally con-

⁸New Jersey Institute of Technology reports capacity of 1,678 in 2012, 11,806 in 2013, and 1,657 in 2014. 2013 capacity is manually imputed as the mean of 2012 and 2014 capacities.

Table 1: Descriptive Statistics for Key Institutional Variables

Variable	Mean	SD	Min	Max	N
Room Capacity	2040.70	2198.16	0.00	19446.00	21344
Total Enrollment	7658.50	9186.95	212.00	96211.00	21347
4-year Graduation Rate (%)	39.47	21.93	0.00	96.00	14521
6-year Graduation Rate (%)	55.85	18.05	0.00	100.00	14521
8-year Graduation Rate (%)	57.88	17.28	0.00	100.00	14521
First-Year Retention Rate (%)	76.44	11.14	16.00	100.00	20248
Beds per Student	0.39	0.25	0.00	3.54	20791
Beds per First-Year	2.34	1.02	0.00	10.59	20300
First Year Cohort Size	1099.62	1249.99	50.00	10099.00	20659
Selectivity Index	1088.61	132.50	564.00	1560.00	18028
On-Campus Room Charge (\$/month)	567.67	237.93	0.00	2604.00	17351
Annual In-State Tuition (\$)	18374.54	13309.90	0.00	58615.00	20913
Instruction Spending (\$mn)	86.11	180.85	0.02	3126.10	20839
Academic Support Spending (\$mn)	23.54	59.80	0.00	1619.43	20839
Student Services Spending (\$mn)	16.69	22.95	0.00	387.40	20839
Scholarships Spending (\$mn)	6.83	16.27	0.00	208.84	20839
Mandatory On-Campus Housing	0.20	0.40	0.00	1.00	20093

Notes:

Sample: IPEDS, 2003-2019. Unit of observation is institution-year. Annual in-state tuition is the published in-state charge, in nominal dollars. Mandatory On-Campus Housing is a binary indicator equal to 1 if the institution requires all first-year students to live on campus, and 0 otherwise.

trolled housing and the associated capacity. Throughout, I use “on-campus housing” as shorthand for institutionally-owned or -controlled beds, which may be located on or near the main campus. Total enrollment ranges from just 212 (Frontier School of Midwifery and Family Nursing in 2004) to 96,211 (Grand Canyon University in 2019), with a mean of 7,659. On average, 76% of first-year students persist to second year, while just 39% of students in 4-year courses graduate within normal time.⁹ Unfortunately, colleges do not report retention rates on a yearly basis after first-year. 76.44% of students proceed to second-year, 39.47% graduate within 4 years, and a further 18.41% graduate in the following four years. 18.56% of students are therefore unaccounted for, and drop out some time after the first-to-second-year transition

⁹Graduation rates are cohort measures; denominators are entering cohorts, not conditional on second-year progression. The graduation rates for this sample following the imposition of the previously outlined sampling restrictions remain in line with national averages reported by the National Center for Education Statistics: <https://nces.ed.gov/FastFacts/display.asp?id=40>.

(or potentially finish their four-year degree after more than eight years).

IPEDS reports capacity (number of beds), but not occupancy or the share of students living in institution-controlled housing. Changes in capacity are therefore shocks to the potential number of on-campus residents. As descriptive context, the average college has enough beds to cater to all first-year students, but significantly fewer beds than the total number of enrolled students.¹⁰ The absolute number of beds in the college (the number of institution-owned beds reported in IPEDS) is used as the measure of capacity in the main analysis, as it represents the cleanest, most exogenous treatment definition: it captures the physical opening (or closure) of dormitory space rather than endogenous shifts in utilization. The beds per student and beds per first-year ratios are not used as treatment definitions, as the estimated effect of an expansion may be confounded by the effect of an increase in total enrollment or first-year cohort size. Instead, total enrollment and incoming first-year cohort size are used as covariates.

The selectivity index is constructed based on test scores of incoming first-year students. As some colleges report SAT scores and others report ACT scores, ACT scores are converted to SAT scores using ACT concordance tables ([ACT Education Corp. 2018](#)). The SAT totals used correspond to the combined verbal and math sections, on a 1600-point scale. A single measure of selectivity is then constructed as the mean of the 25th percentile and 75th percentile of SAT scores if available, and converted ACT scores if not. The index is therefore a proxy for cohort academic performance that varies over time within colleges; this variable captures shifts in entrant quality even when GPA distributions and application essays are unobserved.

Colleges report annual spending in various areas, including instruction, academic support, student services and scholarships; instruction spending is typically the highest. On-campus housing is mandatory for all full-time, first-time undergraduate students in 20% of college-year observations. Such policies have become less common over the sample period; 36.9% of institutions reported having such a policy in 2004, compared to just 3.3% in 2019

¹⁰There are 318 college-year observations in 64 colleges where beds outnumber total enrollment. 57 of these 64 colleges, at some stage in the sample period, have a requirement for all first-year students to live on campus. The phenomenon of beds outnumbering students typically arises after the removal of such a requirement, which is generally followed by a decrease in the number of beds available on campus. Removing these institutions from the sample has no material effect on results.

(although this partly reflects a change in IPEDS reporting instructions in 2012, which tightened the definition of a “mandatory” policy).

To contextualize on-campus room charges in the local housing market, I compare IPEDS annual room charges to local rents measured from the American Community Survey (see Appendix A for full details). The median on-campus room charge over 2015-2019 is approximately \$678 per month. This is below the local median rent for a one-bedroom apartment (\$731 per month), suggesting on-campus housing provides a saving relative to sole-occupancy rental. However, it is substantially above the per-person cost of sharing a two-bedroom unit (\$454 per person per month) - a more realistic off-campus alternative for most students - and at roughly 85% of institutions in the sample, on-campus charges exceed this per-person benchmark.

4.2.1. Descriptive Evidence on Housing Supply Expansions

Figure 1 shows the housing capacity over time for 12 randomly selected institutions in the sample. In a number of cases, the room capacity is steady with some sharp increases. Notably, however, many of the colleges exhibit sharp decreases as well as increases. It is also notable that in many cases, sharp increases in room capacity occur over a period of more than 1 year.

Table 2 presents descriptive statistics regarding which colleges tend to experience changes in capacity, where a change in capacity is defined as an expansion or reduction in total capacity by at least 250 beds in one year. 585 institutions in the sample increased their capacity by 250 beds in one year, while 365 reduced by 250 beds in one year (of these 365, 329 had also had an increase of 250 beds within the sample period, indicating that many reductions in capacity are temporary and followed by corresponding expansions). Of the 585 colleges that expanded, 342 had two or more expansions during the sample period. The average enrollment of the institutions which increased/decreased their capacity over the sample period is higher than those which did not (this is to be expected due to the definition of treatment as an absolute increase/decrease). Colleges which experienced expansions tended to be public (63% of expanding colleges, compared to 20% of non-expanding colleges) and urban (57% of expanding colleges, compared to 41% of non-expanding colleges). There were no differences in the mandatory housing policies for first-years in expanding and non-expanding colleges. Expansions and reductions took place most often in California, Texas and New York, while colleges with static housing supply were most often located in New York, Pennsylvania, and Massachusetts.

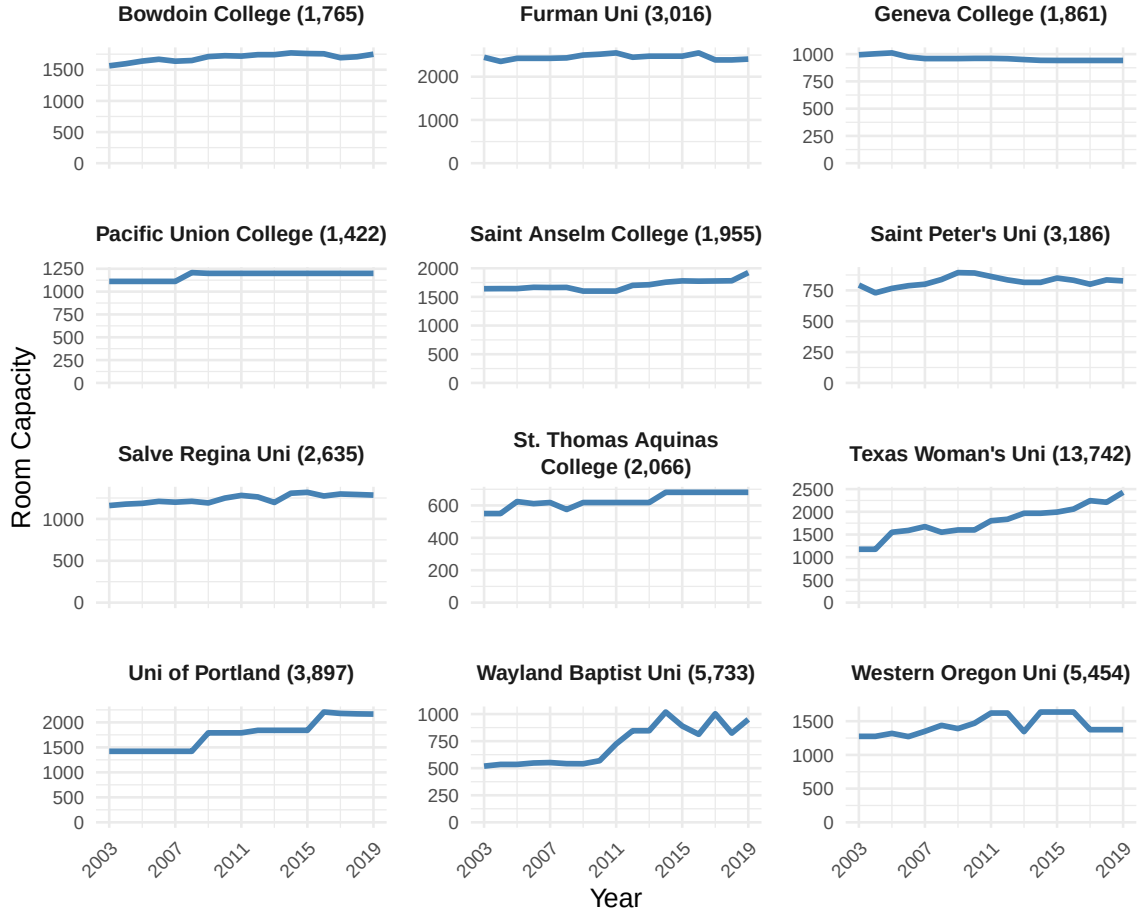


Figure 1: Housing capacity over time for 12 randomly selected institutions. Average enrollment over the sample period shown in parentheses.

Figure 1 and Table 2 make two facts clear. First, capacity paths are, in many cases, erratic. Many colleges experience multiple large increases and decreases. Many expansions are preceded or followed by a contraction of a similar magnitude. These temporary closures may be due to maintenance, or scenarios whereby new dorms are built to allow for the closure of an old facility. Second, treatment propensities are heterogeneous. Colleges that add 250 beds in a single year are, on average, larger, more likely to be public, and more likely to be located in urban areas. These features complicate a conventional staggered difference-in-differences design: units receive treatment

Table 2: Summary of Institutional Characteristics by Expansion/Reduction Status

Treatment Status	Institutional Characteristics					
	Number	Avg Enroll	Public	Urban	Mandatory Residence	Top States
Did Not Expand 250+	671	3296	20%	41%	37%	NY, PA, MA
Expanded 250+	585	12661	63%	57%	36%	CA, TX, NY
Did Not Reduce 250+	891	5169	31%	45%	35%	NY, MA, PA
Reduced 250+	365	13734	63%	56%	42%	CA, NY, TX
Expanded & Reduced	329	14634	64%	56%	42%	CA, NY, TX
Neither Expanded Nor Reduced	635	3171	18%	40%	37%	NY, PA, MA
Expanded 250+ (More than once)	342	16256	70%	60%	38%	CA, TX, NY

Notes:

Table shows descriptive statistics for colleges that expanded (adding 250 beds in a year) or reduced capacity (decreasing by 250 beds). Mandatory residence is defined as requiring first-year students to live on campus in 2004 (the first year in which this information is available).

at different times (and sometimes repeatedly); “never-treated” colleges are structurally different; and the parallel trends assumption is likely to be violated when housing trajectories diverge for structural reasons. It is therefore difficult to clearly label colleges as treated, with a distinct pre-treatment and post-treatment period, and equally challenging to define appropriate control units when their housing capacities also fluctuate significantly. For example, defining treatment as any increase above a given bed-threshold would classify many colleges as treated multiple times, while others just below the cutoff might still undergo substantial, and sometimes sharp, changes in capacity. A simple binary treatment definition fails to capture this complexity and risks misrepresenting the true variation in housing trajectories.

5. Methodology

5.1. Identification Strategy

Because standard two-way-fixed-effects staggered difference-in-differences relies on a single, one-off treatment and the assumption of parallel trends in the absence of treatment, the IPEDS housing data lend themselves poorly to that framework. The primary empirical strategy therefore employs institution-level differencing. Differencing eliminates all time-invariant college characteristics (the chief source of bias given the heterogeneity documented above) and is particularly suited to data of this nature: it exploits the rich within-institution variation (increases, decreases, and periods of stability) without requiring an explicit control group. As a robustness check, a stacked difference-in-differences applied to a restricted subset of colleges with single, discrete capacity jumps is described and estimated in Section 9; this design

validates the main results in a cleaner potential-outcomes setting, but without the large sample size and external validity of analysis based on the entire sample.

There are two endogeneity issues: colleges sorting into treatment, and students sorting into treated colleges. Regarding colleges sorting into treatment, both estimators eliminate all time-invariant differences across colleges by construction. The relevant question for identification is therefore not whether colleges that choose to expand are different from colleges that do not - Table 2 shows that they are - but whether the periods in which housing comes on stream are systematically different from other periods at the same institution in ways that independently affect student outcomes.

The central identification argument rests on the long and largely exogenous production lag between the decision to expand and the delivery of new beds. Building a new residence hall is not a short-run decision: the process typically involves capacity planning, capital budgeting and financing, architectural design and regulatory approval, and a construction phase subject to its own delays. From initial approval to completion, timelines commonly span several years. As a result, conditional on the decision to expand, the exact timing of delivery is unlikely to be systematically related to short-run fluctuations in institutional quality or student composition which could independently affect student outcomes. Identification therefore comes from comparing adjacent cohorts within the same institution, where the expansion decision is typically already fixed but cohorts differ in whether the additional housing has actually come on stream. This isolates variation in realized capacity that is plausibly orthogonal to other short-run shocks.¹¹

Nevertheless, there may be residual concerns about concurrent institutional changes. Housing expansions may coincide with broader initiatives aimed at improving student outcomes, or may crowd out other forms of ex-

¹¹For example, Austin Peay State University’s residence hall expansion - one of the larger capacity increases in the sample - was first outlined in a plan published in the 2006-07 academic year (see https://www.apsu.edu/univ-design-and-construction/master-plan/MasterPlan_2007.pdf), with construction beginning in Fall 2009 and beds, which were initially expected to become available in Fall 2011 (see <https://www.clarksvilleonline.com/2009/03/14/new-dorm-planned-for-apsu-campus/>), eventually coming on stream for the 2013-14 academic year. The seven-year gap between planning and occupancy, including unanticipated delays at the construction stage, illustrates why the cohorts used as the comparators in this study could not have anticipated, let alone responded to, the expansion at the time they enrolled.

penditure. I address these concerns by controlling for changes in spending on instruction, academic support, student services, and scholarships. These controls capture contemporaneous institutional investments.

The second concern is that changes in housing capacity coincide with changes in the scale or composition of enrolled students. Colleges may expand capacity alongside increases in enrollment, or improvements in housing availability may make the institution more attractive to prospective students. I address changes in scale by controlling for cohort size and total enrollment, ensuring that estimates are not driven by mechanical changes in student numbers. Changes in composition are addressed using a selectivity index based on SAT and ACT distributions of entering cohorts. As an additional check, Appendix B reports event studies for a wider set of cohort-composition variables around expansion events, including Pell receipt, gender, race/ethnicity, age, recent high-school graduation, and in-state status. Moreover, given the unpredictable multi-year lag between planning and delivery, it is unlikely that entering cohorts can respond to the precise timing of housing completion. The covariates referenced above are described in detail in Section 5.2, and additional robustness checks are carried out in Section 8.

5.2. Difference Models

I exploit within-college variation by estimating difference models. Let Δ_1 and Δ_4 denote one-year and four-year temporal differences, respectively. Outcomes are measured in percentage points. I begin with a linear benchmark and then allow for non-linearities using bins. For first-to-second-year retention, which is observed after one year, I estimate:¹²

$$\Delta_1 \text{Retention}_{ic} = \alpha + \beta^{(1)} \frac{\Delta_1 B_{i,c}}{100} + \Delta_1 \mathbf{X}'_{ic} \gamma + \Delta_1 \mathbf{Z}'_{ic} \theta + \Delta_1 \varepsilon_{ic} \quad (1)$$

Here $\Delta_1 \text{Retention}_{i,c} = \text{Retention}_{i,c} - \text{Retention}_{i,c-1}$ is the change in the first-year retention rate between cohort c and cohort $c - 1$, and $\Delta_1 B_{ic} = B_{ic} - B_{i,c-1}$ is the change between the number of beds available when cohort c was in first-year and when cohort $c - 1$ was in first-year. The coefficient $\beta^{(1)}$ is interpreted as the percentage-point change in retention per +100 beds.

¹²The intercept α captures a common linear trend in levels; in practice, I also estimate a variant that replaces α with a full set of cohort-year dummies (equivalent to first-differenced year fixed effects), yielding nearly identical results.

Because very small adjustments (and contractions) may have different effects from large, sustained expansions, I also estimate a flexible binned specification:

$$\Delta_1 \text{Retention}_{ic} = \alpha + \sum_j \beta_j^{(1)} \mathbb{I}\{\Delta_1 B_{i,c} \in \mathcal{B}_j\} + \Delta_1 \mathbf{X}'_{ic} \gamma + \Delta_1 \mathbf{Z}'_{ic} \theta + \Delta_1 \varepsilon_{ic} \quad (2)$$

$\mathcal{B} = \{B_1, \dots, B_6\}$ denote six mutually exclusive bins for capacity changes: a large increase ($\Delta_1 B > 750$), a moderate increase ($500 < \Delta_1 B \leq 750$), a small increase ($250 < \Delta_1 B \leq 500$), a small decrease ($-500 \leq \Delta_1 B < -250$), a moderate decrease ($-750 \leq \Delta_1 B < -500$), and a large decrease ($\Delta_1 B < -750$). For each bin j , the model includes the indicator $\mathbb{I}[\Delta_1 B_{i,c} \in \mathcal{B}_j]$; the omitted category is $|\Delta_1 B_{i,c}| \leq 250$. The coefficients $\beta_j^{(1)}$ therefore capture the effect of being in bin j relative to small/no change. The non-linear binning of capacity allows the effect of large expansions or contractions to deviate from that of minor adjustments, a flexibility that would be lost in a purely linear differencing specification.

$\Delta_1 \mathbf{X}'_{i,c}$ contains covariates which control for the possibility that the college may make other changes which affect student outcomes. Specifically, I include the one-year differences in total spending on instruction, academic support, student services and scholarships. $\Delta_1 \mathbf{Z}'_{i,c}$ contains one-year differences in covariates which control for cohort composition effects: the selectivity index constructed based on SAT and ACT scores and the size of the enrolling cohort c .

Standard errors are clustered by college. In the main text, I present two fully specified models - one including the size of the entering first-year cohort and one in which cohort size is replaced by total institutional enrollment (which includes students of all levels) for robustness. Including these covariates in separate specifications avoids potential multicollinearity concerns; results are similar when both are included jointly. Appendices C and D report additional specifications that incrementally add controls.

Graduation is analyzed with four-year differences that match the normal time to degree. Because capacity may influence graduation through multi-year exposure, I define treatment at the cohort level using the average number of beds available over the four academic years a cohort is enrolled. For the cohort entering in year c , exposure is $\bar{B}_{ic} = \frac{1}{4} \sum_{k=0}^3 B_{i,c+k}$. Treatment is the change in this four-year average between cohort c and cohort $c - 4$ in the

same institution ($\Delta_4 \bar{B}_{ic} = \bar{B}_{ic} - \bar{B}_{i,c-4}$). Spending variables enter as four-year differences in cohort-average exposure ($\Delta_4 \bar{X}_{ic}$) to mirror treatment timing, whereas selectivity and cohort size (or total enrollment) are defined at cohort entry and therefore enter as four-year start-to-start differences ($\Delta_4 Z_{ic}$).¹³ The linear benchmark model is:

$$\Delta_4 \text{Graduation}_{ic} = \alpha + \beta^{(4)} \frac{\Delta_4 \bar{B}_{ic}}{100} + \Delta_4 \bar{\mathbf{X}}'_{ic} \gamma + \Delta_4 \mathbf{Z}'_{ic} \theta + \Delta_4 \varepsilon_{ic}. \quad (3)$$

The coefficient $\beta^{(4)}$ is interpreted as the percentage-point change in the four-year graduation rate per +100 average beds over the cohort’s four years. For flexibility, I also estimate the binned analogue, using the same bins as in the retention models:

$$\Delta_4 \text{Graduation}_{ic} = \alpha + \sum_j \beta_j^{(4)} \mathbb{I}\{\Delta_4 \bar{B}_{ic} \in \mathcal{B}_j\} + \Delta_4 \bar{\mathbf{X}}'_{ic} \gamma + \Delta_4 \mathbf{Z}'_{ic} \theta + \Delta_4 \varepsilon_{ic}. \quad (4)$$

Table 3 reports the distribution of $\Delta_1 B_{ic}$ and $\Delta_4 \bar{B}_{ic}$ and the share of observations in each bin.

Standard errors in the graduation models are again clustered at the college level. This design aligns the timing of treatment with the outcome’s horizon: graduation is a four-year outcome, so each cohort’s full four-year exposure enters the comparison. Mid-program expansions or contractions are therefore included in the treated cohort’s average (and in the prior cohort’s average if they occurred earlier), avoiding the attenuation that arises when only entry-year capacity is used. An alternative “start-to-start” specification is estimated in Section 8.3.

Identification relies on the absence of contemporaneous, unobserved shocks that are correlated with discrete capacity changes and with student outcomes. Three potential threats merit discussion. First, contemporaneous

¹³The cohort-average spending measure includes years $c+1$ through $c+3$, which overlap with the period over which the graduation outcome is determined. If housing expansions cause concurrent changes in spending, or if spending responds to intermediate student outcomes, these controls may absorb part of the causal effect or introduce collider bias. An alternative start-to-start specification (Section 8.3), which uses entry-year spending only, provides a robustness check that is free of this concern.

Table 3: Observation counts and percentage shares by capacity-change bin (Graduation and Retention)

Bin	N (Grad)	Share, % (Grad)	N (Ret)	Share, % (Ret)
+ >750	463	5.1	261	1.7
+ 500–750	399	4.4	222	1.4
+ 250–500	1073	11.7	718	4.7
(−250+250, reference)	6869	75.2	13731	88.9
- 500–250	213	2.3	300	1.9
- 750–500	71	0.8	83	0.5
- >750	52	0.6	124	0.8

Note: This table reports the number of institution-year observations falling into each capacity-change bin for both samples. Shares are percentages of total observations within each sample. Positive values indicate increases in housing capacity; negative values indicate decreases.

institutional initiatives that coincide with housing expansions could bias results. This concern is mitigated by the inclusion of detailed spending controls (instruction, academic support, student services, and scholarships), which capture most parallel investments. Second, local shocks may simultaneously affect both construction and outcomes. For example, labor-market changes may influence colleges’ ability to build while also altering students’ opportunity costs of study; zoning or regulatory shifts may affect both the feasibility of new residence halls and the affordability of private rentals. Granular financial and selectivity controls limit the scope for such shocks to drive the results. Third, colleges might build in anticipation of “better” students, or expansions might attract stronger applicants. Both threats seem implausible given the long planning horizons of residence hall projects; residence hall openings typically follow years of design, financing, and construction. Such horizons make it unlikely that expansions respond to the expected quality of a single incoming cohort. Moreover, the selectivity index (based on SAT/ACT scores) flexibly absorbs any shifts in student composition that could bias the estimates. Some further threats remain; for example, a college might increase spending in an area not covered by the controls (for example, by constructing a parallel capital project such as a new student center), or might change its allocation policies around the time of an expansion (for example, prioritizing

weak/strong students for beds). The difference-in-differences analysis in Section 9, which operates in a cleaner causal framework, further supports the robustness of these estimates to potential confounding.

6. Results

6.1. Continuous Difference Models

Table 4: Linear effects per +100 beds

Variable	Retention		Graduation	
	Retention (1)	Retention (2)	Graduation (1)	Graduation (2)
Per +100 beds (pp)	0.001 (0.006)	-0.003 (0.006)	0.112*** (0.029)	0.093*** (0.026)
Controls				
Spending controls	X	X	X	X
Selectivity	X	X	X	X
Cohort size	X		X	
Total enrollment		X		X
Sample size	15,439	15,439	9,140	9,140

Notes:

Coefficients represent the percentage point change associated with a 100-bed capacity increase. Retention uses 1-year differences; graduation uses 4-year cohort-average exposure. Robust SEs clustered by institution; *, **, *** indicate significance at 10%, 5%, and 1%, respectively.

Table 4 reports the results of the fully specified models showing the effects of a 100-bed capacity increase on retention and graduation rates. The unit of observation is the one-year change in retention, or the four-year change (cohort c versus $c - 4$) in graduation. Each coefficient can therefore be interpreted as the percentage point increase in the retention or graduation rate associated with a 100-bed increase in capacity.

Considering first-year retention is measured as an integer between 0 and 100, the point estimates in the first two columns are small. The coefficients are essentially zero and precisely estimated, with confidence intervals that rule out meaningful retention effects. These findings align with the institutional practice of housing virtually all incoming first-years, so adding a few hundred beds rarely alters first-year housing assignments. When expansions occur, any benefits are likely to affect upper years at the margin.

Columns 3 and 4, which present the results of the four-year difference models for graduation outcomes, show a clear positive effect of capacity ex-

pansions on degree completion.¹⁴ An increase of 100 beds raises four-year graduation by roughly 0.1 percentage points, significant at the 1% level. The results are robust to replacing the four-year change in cohort size covariate with four-year change in total enrollment. The full set of specifications in Appendix C show that adding spending, selectivity and cohort size controls does not materially affect the estimates, indicating that the completion effects are not driven by changes in enrollment patterns or institutional spending but by genuine capacity relief.

6.2. Binned Difference Models

The results show that, on average, on-campus housing expansions do not affect first-years but do affect upper-year students across the entire sample. However, there are a number of reasons to suspect that these average effects per 100 beds may not be reflective of the true effects of large expansions. First, expansions and contractions may not have symmetric effects. Second, small incremental changes in capacity, which the descriptive statistics in Section 4.2 show are common, may have minimal effects on overall student outcomes, which would dampen the estimates in the continuous model. Third, such small changes may not represent actual building events, which is the policy lever of interest in the analysis. For this reason, I estimate binned models that capture the effect of discrete capacity expansions and contractions of various sizes, comparing them to periods of little to no capacity changes (absolute change of at most 250 beds). The results for the preferred specification (the full model with the cohort size covariate) are shown in Figure 2.

¹⁴The graduation analysis has fewer observations for three reasons. First, graduation data are only available for cohorts entering through 2015, while retention data extend through 2019. Second, the four-year difference requires non-missing graduation rates in both the focal cohort (t) and the comparison cohort ($t - 4$), which is binding at the start of the panel. Third, all spending controls must be non-missing over both cohort windows, causing additional attrition where finance data are sparse.

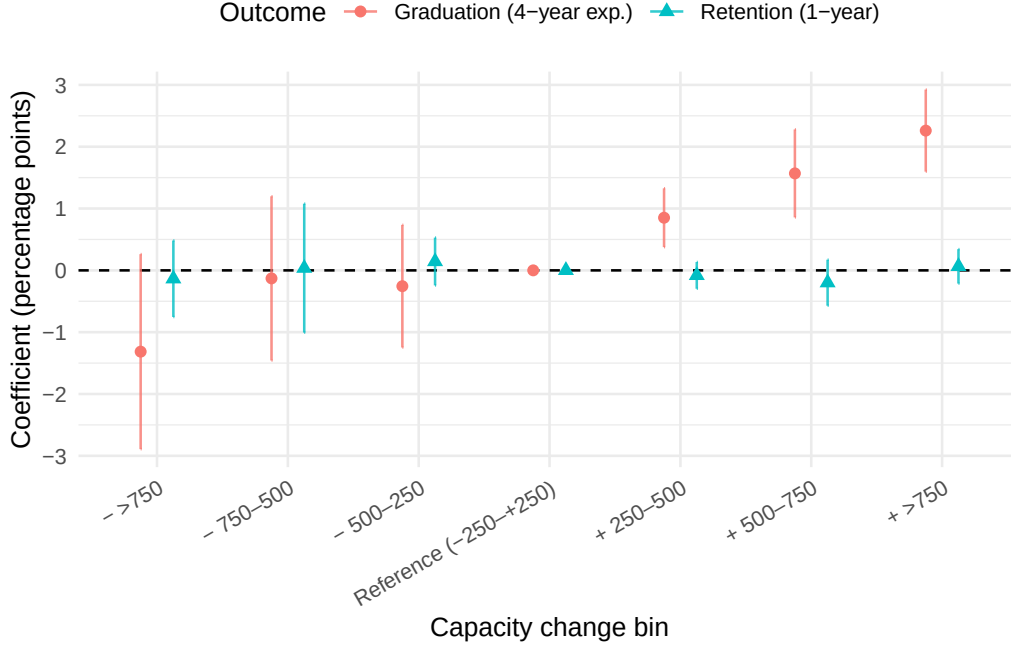


Figure 2: Differencing estimates of the effect of on-campus housing capacity changes on first-year retention and four-year graduation, by size of capacity change. The reference (omitted) category is changes of at most 250 beds. All estimates include controls for institutional spending, selectivity, and cohort size. Points are estimated effects; bars are 95% confidence intervals.

The results of the binned models for retention outcomes are consistent with the continuous model; the estimated coefficients for capacity expansions and contractions of all magnitudes are close to zero. A clearer story emerges regarding graduation effects, however, when bins are used to allow for nonlinearities. First, the effects of contractions in capacity, although directionally negative as expected, are small and statistically indistinguishable from zero. Very large contractions of over 750 beds do show a negative estimate which is large in magnitude, but this coefficient is based on just 52 observations and is not statistically significant. There are a number of possible explanations for this; it may reflect the fact that closures are often temporary and do not affect students for the duration of their time in college. Alternatively, residences are closed because they are in a condition so poor that students do not in fact benefit from living in them. Whatever the reason, it appears

that the effects of dorm closures and openings are not symmetric, and the effect of closures attenuates the results in the continuous model.

The binned model also shows that large expansions have a large and significant effect on graduation outcomes. Increases of 250-500 beds are associated with an increase in graduation rates by roughly 1 percentage point, increases of 500-750 beds are associated with an increase of roughly 1.5 percentage points, and increases of over 750 beds are associated with an increase of over 2 percentage points. These graduation gains are substantially higher than the gains implied by the continuous linear model, and emphasize the importance of allowing for some degree of nonlinearity. The differences can be explained both by the fact that contractions in capacity have practically no effect on outcomes, and by the minor effects of small expansions, which are confined to the reference category in the binned model.

The results show that large discrete expansions in on-campus beds yield large positive effects on four-year graduation rates. This indicates that, even if first-year housing is saturated, boosting supply eases constraints for upper-year students, improving stability and persistence to degree. The retention estimates, meanwhile, imply that incoming cohorts are almost always accommodated, so additional beds rarely alter first-year outcomes. Instead, the graduation gains point to upper-year students as the main beneficiaries: capacity constraints tighten as cohorts progress, and easing them raises completion. The results imply that colleges seeking to improve first-year retention should focus on non-housing factors (such as financial aid, academic support, or orientation), particularly if, as in many colleges, first-years are already sufficiently housed. Colleges aiming to boost graduation rates, on the other hand, may find that targeted investments in upper-year housing capacity pay dividends in timely degree completion.

6.3. Six- and Eight-Year Graduation Rates

An increase in the four-year graduation rate could reflect two distinct effects: students completing on time who would otherwise have finished later (an intensive margin effect on timing), or students completing who would otherwise not have graduated at all. As shown in Table 1, 39% of students graduate within four years, a further 16% complete in years five and six, and a further 2% in years seven and eight. Given the low proportion of students graduating in years seven and eight, it is reasonable to assume that the eight-year rate is close to the practical ceiling for total completions. An effect on the four-year rate that does not extend to the eight-year rate would therefore

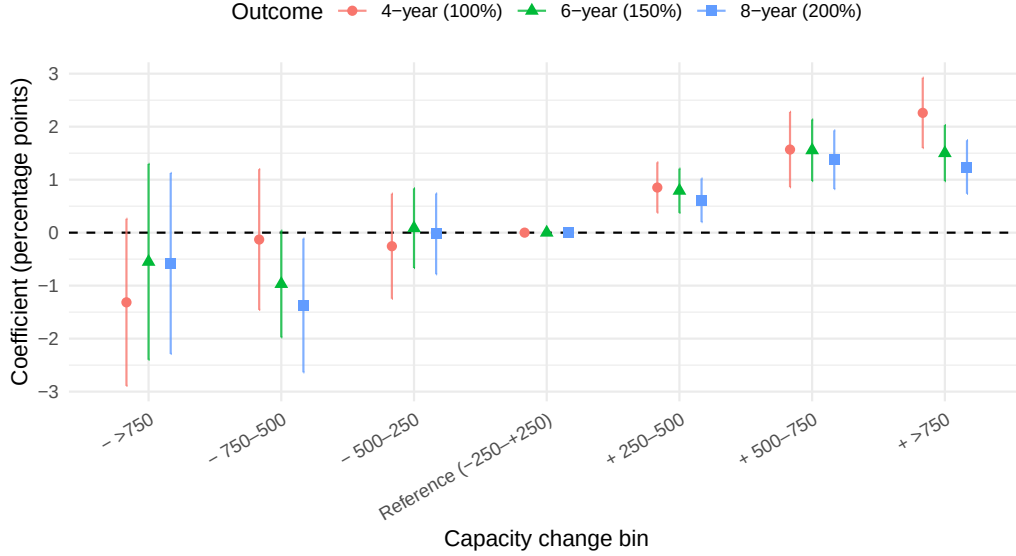


Figure 3: Graduation rate coefficients by capacity-change bin for four-, six-, and eight-year outcomes. All estimates include spending, selectivity, and cohort-size controls. Points are estimated effects; bars are 95% confidence intervals. The omitted category is changes in capacity of at most 250 beds.

indicate that housing expansions accelerate completion rather than increase it; a significant effect at the eight-year horizon would imply genuine gains in total graduation.

To examine this, I repeat the main binned graduation analysis using the six-year (150% of normal time) and eight-year (200% of normal time) completion rates. The specification is otherwise unchanged; I continue to use the average exposure across a student's first four years of enrollment. The six-year rate is also the standard outcome measure used in the IPEDS-based graduation literature, and higher average graduation rates in many other countries make six- and eight-year rates more comparable for international external validity.

As shown in Figure 3, the effects on six- and eight-year graduation rates are similar in magnitude to those on four-year completions, indicating that the main results are driven by students who would not otherwise have graduated, rather than by students graduating more quickly.

7. Mechanisms

Section 2 outlined two channels through which housing expansions may improve student outcomes: a campus residence channel, operating through the direct benefits of on-campus living, and a rent market channel, operating through reduced competition for private rental housing near the campus. Institution-level data do not permit direct observation of students' residential status, so the two channels cannot be separately identified. The analysis therefore tests for evidence of the rent market channel.

Two approaches are taken. The first examines heterogeneity in the main effects across subgroups of institutions - for instance, by local rent levels, urban setting, and the socioeconomic composition of the student body. Systematic differences in the expected directions would be consistent with a rent market mechanism, though such heterogeneity is not definitive, since it may also reflect variation in campus residence effects. The second approach provides a more direct test: if an expansion at one institution reduces local rents, students at nearby institutions should also benefit. Such spillovers are inconsistent with a pure campus residence channel and would constitute direct evidence for the rent market mechanism.

7.1. Subgroup Analysis

Carrying out the analysis across subgroups of institutions provides insight into the mechanisms underlying the main results. I examine heterogeneity by socioeconomic composition, local rents, off-campus housing costs, urban location, sector, room-and-board charges, tuition, in-state enrollment shares, and whether the institution ever required students to live on campus. These dimensions capture variation in students' exposure to local housing markets and in the extent to which on-campus housing may substitute for private rental supply.

If housing expansions operate primarily through a rent market channel, effects should be strongest where students are more exposed to local rental markets and where off-campus housing costs are more salient. This would point to larger effects among colleges in markets where private rents are high relative to on-campus charges, and among colleges serving more economically disadvantaged students. Local rental market pressure is measured using the ACS per-person cost of sharing a two-bedroom unit at the PUMA (Public Use Microdata Area) level (see Appendix A for further details). Other subgroup splits (urban versus rural location, public versus private sector, in-state

enrollment shares, and institutional price levels) provide additional evidence on the institutional settings in which on-campus housing expansions may matter most. Full details on the construction of subgroup variables and the definition of each split are provided in Appendix E.

Figure 4 reports graduation results across these subgroups. Overall, the estimated effects are broadly similar across institutional contexts. There is some variation in magnitudes across panels, but no coherent pattern indicating systematically larger effects in the settings where a rent market channel would be expected to be strongest. The subgroup results therefore do not provide systematic evidence in favor of a rent market channel. Instead, they suggest that the effects of housing expansions are relatively uniform across a broad range of institutional environments.

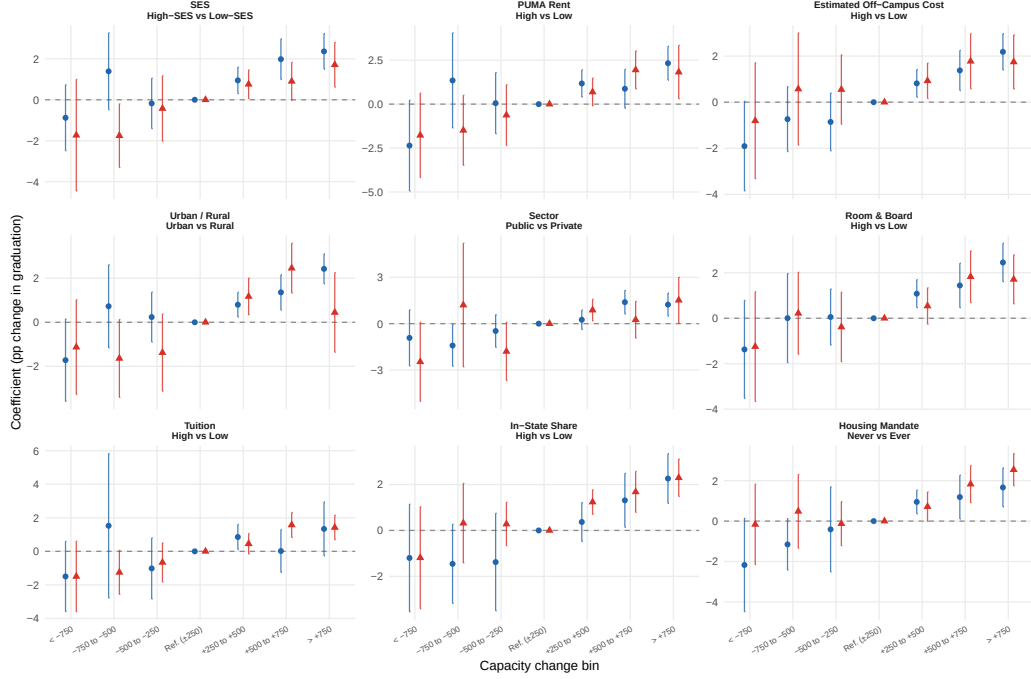


Figure 4: Binned differencing estimates for graduation by subgroup. Within each panel, blue coefficients represent the first-named subgroup and red coefficients represent the second-named subgroup. The Housing Mandate panel compares institutions that never required on-campus residence (Never) with those that did at some point (Ever).

7.2. Spillover Effects

The cleanest test of the rent market channel exploits the geographic structure of local housing markets. If expansions operate partly by reducing local rents, then students at neighboring institutions should also benefit when a focal college builds: the new supply eases pressure in a shared off-campus rental market. Such spillover effects are impossible under a pure residence channel, in which the benefit flows exclusively to students occupying the new beds. The absence of positive spillovers would therefore be difficult to reconcile with a rent market channel. To examine this possibility, I construct a measure of each college’s exposure to neighbor capacity changes and embed it in the first-difference framework used in the main analysis.

Using ZIP5 centroids (latitude/longitude) from the R package “zipcodeR” (Rozzi 2021), I form a geodesic distance matrix across institutions and define, for each college i , the set of neighbors within $R = 20$ miles, $\mathcal{N}_i(R) = \{j \neq i : \text{dist}(i, j) \leq R\}$.¹⁵

For each horizon $h \in \{1, 4\}$ (one year for retention; four years for graduation), let $\Delta_h B_{j,c}$ denote the h -year change in dormitory capacity (or, in the case of graduation, in average capacity over four years) at neighbor j . The “leave-one-out” neighbor exposure for college i in year c is:

$$E_{ic}^{(h)}(R) = \sum_{j \in \mathcal{N}_i(R)} \Delta_h B_{j,c}, \quad h \in \{1, 4\}, \quad R = 20 \text{ miles.} \quad (5)$$

This is zero when no neighbor changes capacity and positive (negative) when neighbors expand (contract). To mirror the main design, I first translate $E_{ic}^{(h)}(R)$ into mutually exclusive neighbor bins.

Let $\Delta_h Y_{ic}$ be the h -year difference in the outcome (for retention, the one-year change in next-cohort retention; for graduation, the four-year change in the 4-year graduation rate). Let $N_{ic}^{(h)}$ collect the neighbor bin indicators, $O_{ic}^{(h)}$ the own capacity bins, and X_{ic} and Z_{ic} the spending and selectivity/size controls, respectively, as in the main models. The binned specification is:

¹⁵Neighbor sets are built once from a full distance matrix over ZIP5 centroids (Haversine distance), avoiding per-row radius searches. Exposure sums are leave-one-out by construction (own capacity is excluded). Spillover regressions include the same differenced controls as the main models. Repeating the exercise with alternative radii (10, 15 and 25 miles) yields the same qualitative conclusion: no robust spillover effects, while own expansions continue to raise graduation.

$$\Delta_h Y_{ic} = \alpha + \beta' N_{ic}^{(h)} + \pi' O_{ic}^{(h)} + \gamma' \Delta_h \mathbf{X}_{ic} + \theta' \Delta_h \mathbf{Z}_{ic} + \Delta_h \varepsilon_{ic}, \quad h \in \{1, 4\}, \quad (6)$$

The spillover models exclude institutions lacking valid ZIP codes (~150 colleges). To improve precision and reduce estimation noise, I collapse the full bin structure into three groups: colleges experiencing a capacity increase of 500 beds or more, colleges experiencing a capacity decrease of 500 beds or more, and those with smaller changes (within ± 500 beds), which serve as the reference category. Standard errors are two-way clustered by institution and ZIP5. Of these colleges, 28.6% have no neighbors within 20 miles, 17.8% have exactly one, 12.9% have two, 6.4% have three, and 34.4% have four or more; institutions with no neighbors are then dropped. If the effects observed in the main results occur primarily through the rent market, we would then expect the estimated coefficients for the neighbor bins to be positive; if effects occur primarily through the campus residence channel, these coefficients will be close to zero.

Table 5: Spillover effects (20 mi): Neighbor capacity changes on retention and graduation

	Retention	Graduation
Capacity Effect		
Neighbor decrease (≥ 500 beds)	0.032 (0.182)	0.975 (1.101)
Neighbor increase (≥ 500 beds)	-0.143 (0.101)	-0.324 (0.244)
Own decrease (≥ 500 beds)	0.239 (0.471)	-0.587 (0.746)
Own increase (≥ 500 beds)	-0.199 (0.185)	1.495*** (0.365)
Controls		
Spending controls	X	X
Selectivity index	X	X
Cohort size	X	X
Sample size		
	9,337	5,542

Notes:

Each column is a first-difference regression using 20-mile neighbor capacity changes.

Treatment bins: decrease of ≥ 500 beds and increase of ≥ 500 beds; omitted: $|\Delta B| < 500$ beds.

SEs two-way clustered by institution and ZIP5; *, **, *** indicate significance at 10%, 5%, and 1%.

The results in Table 5 provide little evidence that neighboring capacity

expansions improve outcomes at nearby institutions. Neighboring expansions have small and statistically insignificant effects on both retention and graduation, while expansions in a college’s own capacity remain positive and significant for graduation despite the smaller sample and the coarser treatment definition. This pattern is difficult to explain if the main graduation gains operate primarily through broad local rent relief, since students at nearby colleges should also benefit from reduced pressure in the shared rental market. It is more consistent with a campus-residence channel, in which the benefits accrue mainly to students who gain access to the additional institutionally controlled beds. As an additional check, Appendix F returns to the main binned graduation specification and adds a direct measure of broader local housing construction: the annual dollar value of building permits for residential buildings with five or more units, measured at the Census Core-Based Statistical Area (CBSA) level, i.e. the local housing market. Adding this measure of private rental-supply growth leaves the estimated effects of own-campus capacity expansions essentially unchanged, and the private-supply measure itself is not economically meaningfully related to graduation outcomes. Overall, these results provide little support for rent-market relief as the primary mechanism. They do not rule out smaller or more localized rental-market effects, but they suggest that the main effects are more likely to operate through access to campus residence.

8. Robustness: Differencing Models

8.1. *Alternative Bins*

The main results from the binned difference models compare periods in which there are small, moderate or large increases and decreases in housing capacity with periods in the same college in which little change occurs. The definitions of “small”, “moderate”, “large” and “little” are somewhat arbitrary, however. For this reason, in Appendix Section G I repeat this analysis using alternative bins. The results are generally consistent with the main models: capacity changes have negligible effects on first-year retention, and capacity expansions have a positive effect on graduation rates, with the magnitude of the effect increasing monotonically with the size of the expansion. The effects of decreases in capacity are not significant at conventional levels. The main results are therefore not an artifact of the chosen cut-offs.

8.2. *Proportional Capacity Changes*

Using absolute changes in beds is useful to capture discrete building events. However, absolute thresholds naturally make large institutions more likely to fall into “expansion” bins and leave smaller ones more often in the reference group. This concentrates treatment among larger colleges and complicates comparisons across very different baselines. For example, the effect of a 250-bed increase is likely to vary based on whether prior capacity was 250 or 2,500. As a robustness check, I repeat the binned analysis using proportional rather than absolute capacity changes, under two symmetric binning schemes (5-10% and 10-20% thresholds). Results are consistent with the main findings: retention effects are statistically indistinguishable from zero across both schemes, and capacity expansions deliver positive, significant gains in graduation rates. Full details of the specification and results are reported in Appendix [H](#).

8.3. *Start-to-Start Difference Models*

Although the main specification aligns treatment with a cohort’s full four-year exposure, it is informative to verify that results are not driven by the averaging window itself. A “start-to-start” design (comparing capacity at entry for adjacent cohorts) cleanly compares the housing environment students in different cohorts face when they arrive. This “start-to-start” definition implicitly assumes that the path of capacity in the intervening years (years 2–4 of enrollment) does not affect graduation. Aside from the use of first-year measures of capacity and spending controls, the model is equivalent to the main analysis. The results in Appendix [I](#) are consistent with the main findings: increases in entry-year capacity have a statistically significant effect on graduation rates. The slightly smaller magnitudes relative to the main results are expected, as the start-to-start design: (i) does not credit post-entry expansions that help the focal cohort, and (ii) does not net out pre-entry expansions that helped the previous cohort, attenuating the estimates.

8.4. *Count Model*

The main analysis models graduation rates; an alternative is to model the number of graduates directly. Since each additional bed can accommodate at most one additional student, effects on counts provide a complementary interpretation in units of the treatment. In this section, I replace the outcome with the four-year change in the number of four-year graduates and measure treatment in raw beds rather than per-100 units, so the coefficient

is additional graduates per additional average bed. Cohort size and total enrollment enter as controls to ensure that any increase in graduate counts is not driven mechanically by enrollment growth; all other aspects of the specification are unchanged.

Results are shown in Appendix J. The coefficient is positive and significant across all specifications, consistent with the rate-based results. The fully specified estimate implies approximately 0.05 additional four-year graduates per additional average bed, or about five additional graduates per 100 added beds. This should be read as a reduced-form capacity effect: the additional beds need not all be occupied by students whose graduation outcomes are affected, and colleges may allocate new capacity across cohorts and student years in different ways. The count estimate therefore captures the number of additional on-time completions generated by expanding capacity, rather than an individual-level effect of receiving an on-campus bed.

9. Difference-in-Differences

9.1. Design

As a further check on the differencing results, I implement a cohort-level stacked difference-in-differences design that exploits large, discrete expansions in housing capacity. Rather than hand-curate events, I apply transparent selection rules to each college’s capacity series. An expansion in year t is included as a treatment event if the following simultaneous criteria hold:

1. Large, discrete increase: a one-year increase of at least 300 beds, where the increase is at least 10% of capacity in $t - 1$ and at least 2.5% of total enrollment in $t - 1$.
2. Stable pre-period: in each of the four prior years $t - 4, \dots, t - 1$, there are no annual proportional changes greater than 5%.
3. No immediate reversals: in $t, \dots, t + 3$ there are no contractions of over 150 beds or 5%.
4. One event per institution: if multiple years in one college satisfy conditions 1-3, I retain the first qualifying event.

Robustness to different selection criteria is examined in Appendix Section K. Applying these rules yields 51 treatment events for retention and 46 for graduation. The smaller number for graduation reflects two constraints: the use of four-year differences, which reduces usable observations at the start of the sample, and the availability of graduation data only through 2015, which

truncates the end of the sample. The treatment events are listed and plotted in Appendix L. The intent of this selection is to focus on plausibly exogenous large shocks to on-campus supply, i.e., sharp, isolated expansions that are less likely to be the mechanical by-product of gradual institutional change, and which are large enough to affect student outcomes through on-campus availability and potentially local rent markets.

For each treated college event (i, t) , I construct a two-period panel for each outcome. For graduation, the comparison is between students entering in the year of the expansion (exposed throughout their four years of study) and those entering four years earlier (unexposed throughout). For retention, the comparison is between students entering in the expansion year (exposed in first year) and those entering one year earlier (unexposed in first year). In both cases, later cohorts are excluded. The same stacked difference-in-differences framework is applied to both outcomes.

I add control rows from colleges that did not experience treatment events (as defined in points 1-4 above). Controls are eligible only if they exhibit eight consecutive years of stable capacity (no annual change exceeding 200 beds or 5%). For each treatment event, I include all such colleges with stable capacity over the corresponding event window (from $t - 4$ to $t + 3$). The resulting stacked panel contains, for each event, two rows for the treated college (pre-treatment and post-treatment cohorts) and, for each control college, two rows in the same calendar years.¹⁶ I estimate the following difference-in-differences model:

$$Y_{ic} = \delta(Treated \times Post)_{ic} + \mathbf{X}'_{ic}\gamma + \mathbf{Z}'_{ic}\theta + \alpha_i + \lambda_c + \varepsilon_{ic} \quad (7)$$

Where $(Treated \times Post)_{ic} = 1$ for the treated cohort row and 0 otherwise,

¹⁶To construct the final stacked panel, each event window contributes two years of data per unit (pre-treatment and post-treatment cohort). Because some never-treated colleges appear as controls for multiple events, I compute the number of times each college-year appears and weight observations by the inverse of that count ($w_{it} = 1/n_{it}$), where n_{it} is the number of times that observation appears across treatment stacks). I then collapse the stacked data to a single row per college-year, taking weighted means of the outcome and all covariates using these weights. This ensures that each college-year contributes equally to the estimation, regardless of how often it is reused as a control. The regressions are then estimated on this collapsed dataset. Collapsing the weighted stack to one row per college-year is algebraically equivalent to running weighted least squares on the full stacked dataset with inverse-frequency weights; it is a computational shortcut only.

α_i are college fixed effects, λ_c are cohort (year) fixed effects, and $\mathbf{X}'_{ic}$ and $\mathbf{Z}'_{ic}$ are vectors containing the same covariates as in the differencing regressions. Standard errors are clustered by college.

The coefficient δ compares the average four-year graduation rate (or first-to-second-year retention rate) of the first cohort exposed to the capacity jump with that of an earlier cohort at the same college, net of shocks which affect the treated and never-treated colleges equally. Because only two cohort rows are retained per treated college, this design is not subject to the heterogeneous-timing bias that plagues conventional two-way fixed-effects difference-in-differences with staggered adoption. Identification rests on the parallel trends assumption: that treated and control institutions would have followed similar outcome trajectories in the absence of housing expansions. To make this as credible as possible, I match each treated institution to its K nearest control institutions ($K = 10$) based on pre-treatment outcome trends and observable characteristics.¹⁷ Specifically, for each treatment event, I calculate a vector of pre-treatment covariates over the four years prior to treatment. These covariates comprise the level, mean, and estimated linear trend of the relevant outcome variable; baseline housing capacity (measured in the final pre-treatment year); the average number of beds per first-year student across the four-year window; and averages of cohort size, selectivity, scholarships, and spending on instruction, academic support, and student services. An identical set of variables is computed for all institutions eligible to serve as controls (i.e., those with stable housing capacity in the surrounding period).

For each treatment event, the K nearest control institutions are selected by standardised Euclidean distance on the pre-treatment covariate vector, independently for each event and outcome, and the resulting matched lists are used to construct the stacked panels.

To assess the plausibility of parallel trends on the matched sample, I estimate event-study models on cohorts in the six years prior to each expansion. Figure 5 presents the results.¹⁸ Pre-treatment coefficients are small

¹⁷To ensure that results are not sensitive to the number of matches for each treatment event, I report estimates across different values of K in Appendix M. Varying the number of matched controls K leaves the difference-in-differences estimates essentially unchanged, indicating that the results are not an artifact of a particular matching width.

¹⁸Without matching, point estimates are larger but the graduation event study exhibits a visible upward drift in the pre-treatment period: treated institutions have rising graduation rates relative to the full control pool even before expansion. Although this drift is not

and statistically insignificant for both retention and graduation, supporting the parallel trends assumption.

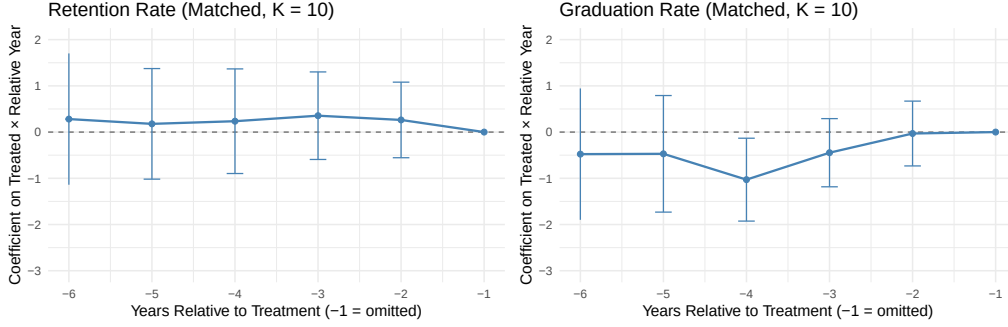


Figure 5: Pre-treatment event studies on the $K = 10$ matched sample. Retention (left) and graduation (right). Year -1 is the omitted reference; bars are 95% confidence intervals.

9.2. Results

Table 6: Difference-in-Differences Estimates (Matched Sample, $K = 10$)

	Retention Models		Graduation Models	
	(1) Retention	(2) Retention + Controls	(3) Graduation	(4) Graduation + Controls
Post $\times$ Treated	0.346(0.392)	0.526(0.402)	1.187*(0.637)	1.416**(0.604)
Cohort Size		X		X
Selectivity Index		X		X
Scholarships		X		X
Instruction Spend		X		X
Academic Support		X		X
Student Services		X		X
Observations	783	783	757	757
Treatment Events	51	51	46	46
Unique Control Colleges	215	215	192	192

^a Stacked difference-in-differences estimates of the effect of large on-campus housing expansions on first-year retention and four-year graduation rates. Estimates based on matched samples ($K = 10$). Institution and year fixed effects in all columns. Controls as indicated. Robust standard errors clustered by institution in parentheses.

Table 6 reports difference-in-differences estimates for retention and graduation. For first-year retention, point estimates are positive but imprecisely estimated: 0.346 without controls and 0.526 with controls, where neither

statistically significant (except marginally at $t=4$), it suggests that treated and unmatched control institutions may differ systematically. Matched estimates are therefore preferred as the more conservative baseline.

estimate is statistically significant at conventional levels. As in the differencing models, this is consistent with a scenario in which first-year students are typically housed on campus, so upper-year students benefit at the margin in the event of an expansion.

Capacity expansions do have a clear positive effect on four-year graduation rates. After the inclusion of all controls, an expansion increases the graduation rate by 1.4 percentage points. The point estimate increases slightly after adding covariates, indicating that observable cohort and spending characteristics were, if anything, less favorable in expansion years. The magnitude is similar to the estimated effects of increasing capacity by over 500 beds in the binned differencing models.

While these estimates stem from a relatively small subset of discrete treatment events,¹⁹ they offer a cleaner causal contrast by leveraging sharp capacity jumps and flat-capacity control windows. Despite its narrower scope, the binary design delivers a sharply identified estimate of the causal impact of a major housing expansion. Reassuringly, these results reinforce the main findings of the differencing approach: housing expansions lead to sizable improvements in graduation outcomes, with no detectable average effect on first-to-second-year persistence.

9.3. Treatment Definitions

To ensure that the results are not driven by the particular treatment definition, I re-estimate the design under four alternative sets of treatment thresholds, keeping all other elements fixed. Specifications 1 through 5 range from the most restrictive (9 events) to the most permissive (117 events), with Specification 3 as the main analysis (46 events); full threshold details are reported alongside the results in Appendix K. Graduation estimates are positive in all five specifications and statistically significant in all but the smallest sample. Point estimates are stable across Specifications 2 through 5 (roughly 1.4 to 1.6 percentage points); the loss of significance in Specification 1 (9 events)

¹⁹“Observations” report unique institution–year rows in the stacked panel after collapsing duplicates that arise when the same control institution–year is matched to multiple events. The retention design (using the short $(t - 1, t)$ window) reuses control years more heavily than the graduation design (using $(t - 4, t)$), so more duplicate rows are collapsed. This explains the somewhat counterintuitive result that the retention model has a smaller sample size than the graduation model, despite the fact that retention has more treatment events and more controls.

and the marginal significance in Specification 2 (31 events) reflect limited sample size. The result is present whenever the sample is large enough to detect it. As a further check on the stability of the main estimate, re-estimating the model after dropping each of the 46 graduation treatment events one at a time yields point estimates ranging from 1.18 to 1.58 percentage points, all significant at the 5% level, confirming that no single institution drives the result (Appendix N).

9.4. *Exposure to the Expansion*

The differencing models in Section 6 measure housing supply for a given cohort as average capacity over the four years of the degree, so a cohort fully exposed to an expansion throughout all four years benefits more than one exposed for only part of its degree. An advantage of the difference-in-differences design is that this exposure gradient can be examined directly. The main estimate compares the first cohort to spend all four years under the expanded capacity (entering at t) to a fully pre-expansion cohort ($t - 4$). Cohorts already enrolled when the expansion occurred - those entering at $t - 1$, $t - 2$, and $t - 3$ - received progressively less exposure: one, two, or three of their four degree-years elapsed before additional capacity was available. If the graduation benefit scales with duration of exposure, the effect should attenuate monotonically as the treated cohort is shifted backward toward the expansion year. To test this, I re-estimate the matched graduation difference-in-differences comparing each already-enrolled cohort ($t - 1$, $t - 2$, $t - 3$) against the same fully pre-expansion cohort ($t - 4$) used in the main analysis.

Figure 6 presents the results. The estimated graduation effects are positive and attenuate monotonically, consistent with an exposure mechanism: cohorts that spend a larger share of their degree under the expanded on-campus environment benefit more. For students one year into their degree at the time of the expansion ($t - 1$), the effect is approximately 1.19 percentage points. For those two years in ($t - 2$), the effect falls to roughly 1.11 percentage points, and by three years in ($t - 3$), to 0.76 percentage points. The monotonic gradient is also reassuring for identification: students who had already enrolled when the expansion occurred could not have chosen the institution on the basis of the new capacity, and any concurrent institutional initiatives aimed at improving outcomes would most plausibly target incoming rather than continuing students. The results therefore hold for cohorts that are largely insulated from both student-selection and co-investment confounds.

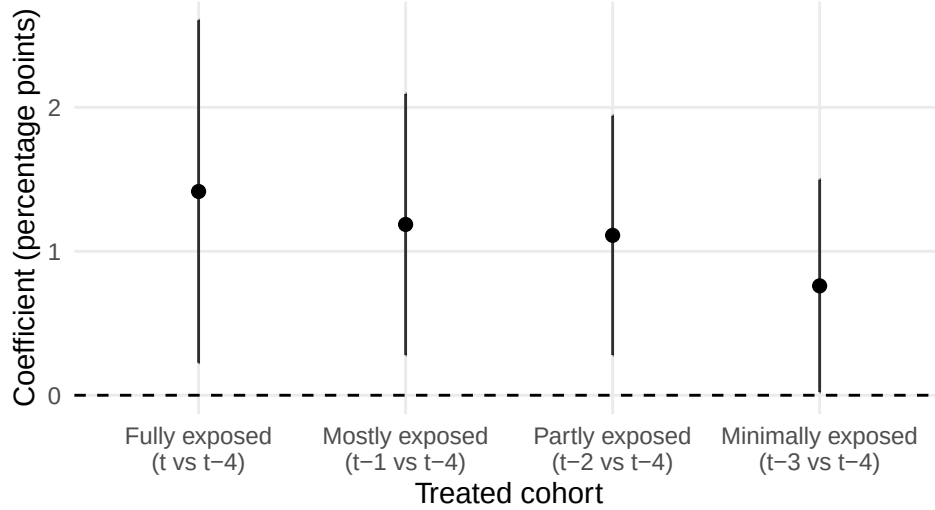


Figure 6: Graduation effect by treated cohort. The first point (t) is the main difference-in-differences estimate (first fully-exposed cohort vs $t - 4$); subsequent points shift the treated cohort to students already enrolled one, two, and three years before the expansion. All specifications include controls. Points are estimated effects; bars are 95% confidence intervals.

10. Discussion

The national differencing design yields a coherent story about how on-campus housing supply influences student success, a pattern confirmed by the stacked difference-in-differences in Section 9. In the differencing models, even large absolute swings in reported beds barely move first-to-second-year retention. Similarly, in the difference-in-differences that isolates discrete, clearly timed expansions and compares the first exposed cohort with its immediate predecessor, the retention estimates are positive but imprecise. The natural explanation is that most expansions occur where first-years were already prioritized for housing, leaving little margin for improvements in retention. Although I do not observe within-college housing allocation directly, this interpretation is consistent with institutional housing policies and aggregate capacity patterns: the vast majority of colleges state that first-year students receive priority, guarantees, or designated housing, the average college in the sample has 2.34 beds per first-year student, and beds exceed first-year cohort size in 90% of college-year observations.

Completion is more responsive than the first-year transition. Across difference models, four-year graduation rates improve meaningfully after siz-

able expansions: while the average effect per +100 beds is about 0.1 percentage points, large expansions of over 250 beds are associated with improvements of 1-2.25 percentage points depending on the magnitude of the expansion. The difference-in-differences estimates are consistent with these findings; large expansions increase graduation rates by approximately 1.4 percentage points. Because baseline four-year graduation averages ~40%, these effect sizes, though numerically modest, are educationally consequential. For a college enrolling 2,500 first-years, a 2 percentage point improvement in on-time completion implies approximately 50 additional degrees per cohort.

Effects are larger for cohorts with longer exposure to housing expansions. They reflect increases in total completion rather than a re-timing of degrees. Overall, the evidence is consistent with an on-campus residence channel rather than a rent market mechanism.

The findings have key implications for college administrators. Large, discrete housing expansions are a completion-enhancing investment. Even in the absence of retention gains, the accumulated improvements in graduation imply dozens of additional degrees per typical cohort at medium-to-large institutions. Where short-run persistence is a priority, institutions may wish to pair capacity policy with targeted supports (orientation, advising, financial aid, or academic support) rather than relying on beds alone, particularly when first-years are largely already housed.

The two channels emphasized in the framework imply that allocation interacts with institutional objectives. Where first-year housing is not universal, prioritizing entrants is the configuration most likely to influence the first-year transition; where the objective is on-time completion, allocating some capacity to later-year students is most likely to influence persistence beyond the first year. The effectiveness of expansions likely depends on how new capacity is allocated across student years, underscoring the importance of institutional housing policies. The following back-of-the-envelope calculations illustrate the cost per additional graduate from both the college's and the state's perspective.

From the college's perspective, residence halls are typically operated as auxiliary enterprises, meaning that operating costs and capital charges are largely covered by room revenue.²⁰ If housing is fully self-financing, the

²⁰In U.S. higher-education reporting, auxiliary enterprises (e.g., residence halls, dining) are defined as activities managed to be essentially self-supporting from user fees

marginal cost to the central academic budget is close to zero. Let $s \geq 0$ denote the net subsidy per bed-year from the central budget (after accounting for room revenue), B^* the number of new beds, Δg the graduation-rate effect in percentage points, and N the size of a typical entering cohort. The annual central-budget cost per additional graduate, once the expansion is in place, is $s \times B^* / [(\Delta g / 100) \times N]$.

When $s \approx \$0$, the central-budget cost per additional graduate is correspondingly close to zero. Even when a subsidy is required, the implied cost per graduate remains modest relative to typical higher-education expenditures. Housing is a durable intervention: once built, additional capacity can benefit each entering cohort until the asset is retired. Unlike many student-success programs that require repeated annual funding, a single capital investment can affect completion for many future cohorts.

A separate calculation is relevant from the perspective of the state, which may finance construction through one-off capital grants. If the state provides a grant of size G and the expansion remains in service for L entering cohorts, the gross public capital cost per additional graduate over the asset's life is $G / [L \times (\Delta g / 100) \times N]$, ignoring discounting. This is a gross state cost, not a net institutional cost: it does not subtract any future recovery of costs through room revenue. Because housing is a durable investment, the cost is spread over many cohorts, substantially lowering the cost per additional graduate.

To illustrate magnitudes, consider a stylized example of a 750-bed expansion at a college enrolling $N = 2,500$ first-year students. A conservative estimate of the effect of such an expansion is a 1.55 percentage point increase in graduation; this is in line with the effect of a 500-750-bed expansion in the binned differencing model; it is also roughly in line with the difference-in-differences coefficient. A 1.55 percentage point increase implies approximately 39 additional graduates per cohort.

From the college's perspective, the annual central-budget cost per additional graduate depends on the net subsidy per bed-year, s . It is \$0 when $s = \$0$, rising to \$19,230 when $s = \$1,000$ and \$48,076 when $s = \$2,500$.

From the state's perspective, the relevant margin is the one-off capital

(see IPEDS glossary: <https://surveys.nces.ed.gov/ipeds/public/glossary>); several state systems specify auxiliaries operate on a self-supported basis (for example, the University System of Georgia: https://www.usg.edu/business_procedures_manual/section15/manual/C1492).

cost. If the expansion remains in service for 30 cohorts, it generates approximately $30 \times 39 = 1,170$ additional graduates. A \$10 million capital grant would therefore amount to roughly $\$10,000,000 / 1,170 \approx \$8,547$ per additional graduate over the asset’s life, before discounting. These calculations are illustrative: they depend on a single point estimate, assume a stable treatment effect across cohorts, and ignore equilibrium responses. Nevertheless, these figures suggest that housing expansions can generate additional graduates at relatively low cost, particularly when room revenue covers operating expenses and capital costs are spread over many cohorts.

The above calculations capture only the fiscal costs of expansion. They do not account for fiscal returns to the state from additional degree completion. Quasi-experimental evidence suggests that degree completion results in substantial labor-market gains for marginal students. Zimmerman (2014) finds earnings gains of 22 percent between 8 and 14 years after high-school completion, while Denning et al. (2019) note that in their setting bachelor’s degree receipt is associated with roughly \$32,000 in additional annual earnings. Taking a deliberately conservative calibration of \$30,000 in additional annual earnings and a 20 percent effective tax rate implies about \$6,000 in additional annual tax revenue per additional graduate. Under the calibration above, one newly graduating exposed cohort would therefore generate about $39 \times \$6,000 = \$234,000$ in additional annual tax revenue while that cohort remains in the labor market; as successive exposed cohorts graduate, these annual tax revenues accumulate across cohorts. These fiscal returns are only one component of the broader social return, since most of the earnings gains accrue privately to graduates.

The policy implication is therefore twofold. First, for colleges that operate housing on an auxiliary basis, expansion may be close to self-financing if room revenue covers operating and capital costs. Second, even if a state subsidizes construction and does not capture rents directly, housing can still be a comparatively inexpensive way of generating additional graduates because one capital intervention can benefit many future cohorts. These figures are illustrative and institution-specific, but they suggest that housing expansions can be a cost-effective route to higher completion both as a low-net-cost institutional investment and as a durable public capital investment.

However, as the evidence points to the campus residence channel as the primary driver of improved outcomes, an important question for policy is what specific aspects of on-campus living generate these gains. While residence halls can generate positive returns for institutions, they require sub-

stantial upfront capital investment and may be difficult to finance. Expanding capacity therefore may not always be feasible. If the benefits derive less from the sheer number of beds and more from the environment they facilitate (peer networks, proximity to services, or on-campus support), then investments that replicate or enhance these features, even without large construction projects, could deliver comparable benefits at lower upfront cost.

Because dormitories are often profit-generating enterprises for colleges, two additional college-side mechanisms could also explain the effects. First, rental-revenue incentives: after an expansion, institutions may have stronger incentives to retain students who pay for institutional housing, leading to institutional practices that raise persistence or completion. Second, performance justification: institutions may intensify completion efforts to demonstrate returns on capital spending (and maintain donor/political support), mechanically lifting graduation rates. Future research with data on precise grades, and not just coarse measures of retention and graduation, could seek to examine whether colleges appear to manipulate grades (for instance through changes in grade dispersion or pass-rate thresholds) following expansions.

11. Conclusion

This paper shows that expanding on-campus housing capacity improves student success. In a differencing design spanning 17 years and 1,256 colleges, large capacity increases have no average effect on first-year retention but raise four-year graduation by about 1-2.25 percentage points, with effects increasing in the size of the expansion. A rule-based difference-in-differences that isolates discrete expansions at large colleges finds an approximately 1.4-percentage-point gain in graduation and no precise evidence of retention effects.

The mechanism analysis provides additional insight into how housing expansions affect completion. Comparing cohorts with differing lengths of exposure to an expansion suggests that longer exposure is associated with larger graduation gains. Extending the outcome window to six- and eight-year graduation rates indicates that these effects reflect increases in overall completion rather than simply a re-timing of degrees.

Evidence on the underlying channel points away from a rent market mechanism. Subgroup analyses do not reveal systematic differences in effects across institutional characteristics, including local rent conditions or student

socioeconomic composition. Consistent with this, there is no evidence of spillover effects on neighboring colleges, and local private multi-family construction does not predict graduation outcomes. These findings suggest that the effects are unlikely to operate through broader changes in local housing markets and are more consistent with improvements in access to on-campus housing.

Large, discrete expansions are completion-enhancing investments; where first-years are already largely housed, institutions seeking immediate persistence gains should focus on targeted supports, while broader expansions can raise on-time graduation. Overall, relaxing on-campus housing constraints stabilizes enrollment through to degree. Future work linking capacity expansions to student-level survey data would help identify which aspects of campus living drive the observed effects and enable analysis of more detailed academic outcomes (such as GPA) and non-academic measures (including student well-being).

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Data Availability

The IPEDS, ACS, and BPS data used in this paper are publicly available from the respective sources cited in Section 4. The analysis code and constructed datasets are available from the corresponding author upon request.

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Declaration of Competing Interests

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- ACT Education Corp. 2018. *ACT / SAT Concordance*. <https://www.act.org/content/act/en/products-and-services/the-act/scores/act-sat-concordance.html>.
- Angrist, Joshua, Daniel Lang, and Philip Oreopoulos. 2009. "Incentives and Services for College Achievement: Evidence from a Randomized Trial." *American Economic Journal: Applied Economics* 1 (1): 136–63. <https://doi.org/10.1257/app.1.1.136>.
- Araujo, Pedro de, and James Murray. 2010. *Estimating the Effects of Dormitory Living on Student Performance*. <https://dx.doi.org/10.2139/ssrn.1555892>.
- Arcidiacono, Peter. 2004. "Ability Sorting and the Returns to College Major." *Journal of Econometrics* 121 (1): 343–75. <https://doi.org/10.1016/j.jeconom.2003.10.010>.
- Arcidiacono, Peter, Patrick Bayer, and Aurel Hizmo. 2010. "Beyond Signaling and Human Capital: Education and the Revelation of Ability." *American Economic Journal: Applied Economics* 2 (4): 76–104. <https://doi.org/10.1257/app.2.4.76>.
- Blimling, Gregory S. 1989. "A Meta-Analysis of the Influence of College Residence Halls on Academic Performance." *Journal of College Student Development*.
- Bozick, Robert. 2007. "Making It Through the First Year of College: The Role of Students' Economic Resources, Employment, and Living Arrangements." *Sociology of Education* 80 (3): 261–85. <https://doi.org/10.1177/003804070708000304>.
- Denning, Jeffrey T., Benjamin M. Marx, and Lesley J. Turner. 2019. "Propelled: The Effects of Grants on Graduation, Earnings, and Welfare." *American Economic Journal: Applied Economics* 11 (3): 193–224. <https://doi.org/10.1257/app.20180100>.
- Dobkin, Carlos, Ricard Gil, and Justin Marion. 2010. "Skipping Class in College and Exam Performance: Evidence from a Regression Discontinuity Classroom Experiment." *Economics of Education Review* 29 (4):

- 566–75. <https://doi.org/10.1016/j.econedurev.2009.09.004>.
- Guzman, Matthew. 2021. “The Effects of on-Campus Housing Requirements on Student Outcomes.” <https://web.archive.org/web/20250804154546/https://www.matthew-guzman.com/wp-content/uploads/2021/12/20211207-Guzman-EAD942-Final-Paper.pdf>.
- Hauschildt, Kristina, Christoph Gwosc, Hendrik Schirmer, Sylvia Mandl, and Cordelia Menz. 2024. *Social and Economic Conditions of Student Life in Europe: Eurostudent 8 Synopsis of Indicators 2021-2024*. Wbv Media GmbH & Company KG. <https://doi.org/10.3278/6001920ew009>.
- Lopez Turley, Ruth N., and Geoffrey Wodtke. 2010. “College Residence and Academic Performance: Who Benefits from Living on Campus?” *Urban Education* 45 (4): 506–32. <https://doi.org/10.1177/0042085910372351>.
- National Center for Education Statistics. 2014. *National Postsecondary Student Aid Study: 2012 Undergraduate (NPSAS:12)*. <https://nces.ed.gov/surveys/npsas/>.
- National Center for Education Statistics. 2026. *Integrated Postsecondary Education Data System (IPEDS)*. U.S. Department of Education. <https://nces.ed.gov/ipeds/>.
- Oreopoulos, Philip, and Kjell G. Salvanes. 2011. “Priceless: The Nonpecuniary Benefits of Schooling.” *Journal of Economic Perspectives* 25 (1): 159–84. <https://doi.org/10.1257/jep.25.1.159>.
- Perez-Felkner, Lara, Christine Baker-Smith, and Sara Goldrick-Rab. 2022. *Does Rent-Free Community Housing Make a Difference in Higher Education Outcomes?* Hope Center for College, Community, and Justice.
- Reynolds, C. Lockwood. 2020. “The Effect of Dormitory Residence During College on Student Outcomes.” *Journal of Human Capital* 14 (2): 249–89. <https://doi.org/10.1086/709534>.
- Rozzi, Gavin. 2021. *zipcodeR: Data and Functions for Working with US ZIP Codes*. <https://CRAN.R-project.org/package=zipcodeR>.
- Sacerdote, Bruce. 2001. “Peer Effects with Random Assignment: Results for Dartmouth Roommates.” *The Quarterly Journal of Economics* 116 (2): 681–704. <https://doi.org/10.1162/00335530151144131>.
- Schudde, Lauren T. 2011. “The Causal Effect of Campus Residency on College Student Retention.” *The Review of Higher Education* 34 (4): 581–610. <https://dx.doi.org/10.1353/rhe.2011.0023>.
- Stinebrickner, Ralph, and Todd R. Stinebrickner. 2006. “What Can Be Learned about Peer Effects Using College Roommates? Evidence from New Survey Data and Students from Disadvantaged Backgrounds.” *Journal of Public Economics* 90 (8): 1435–54. <https://doi.org/10.1016/>

- [j.jpubeco.2006.03.002](#).
- Thompson, Jane, Virginia Samiratedu, and John Rafter. 1993. “The Effects of on-Campus Residence on First-Time College Students.” *NASPA Journal* 31 (1): 41–47. <https://doi.org/10.1080/00220973.1993.11072336>.
- U.S. Census Bureau. 2019. *TIGER/Line Shapefiles*. U.S. Census Bureau. <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>.
- U.S. Census Bureau. 2020. *American Community Survey 5-Year Estimates*. U.S. Census Bureau. <https://www.census.gov/programs-surveys/acs>.
- U.S. Census Bureau. 2021. *Building Permits Survey*. U.S. Census Bureau. <https://www.census.gov/construction/bps/>.
- Webber, Douglas A., and Ronald G. Ehrenberg. 2010. “Do Expenditures Other Than Instructional Expenditures Affect Graduation and Persistence Rates in American Higher Education?” *Economics of Education Review* 29 (6): 947–58. <https://doi.org/10.1016/j.econedurev.2010.04.006>.
- Zimmerman, Seth D. 2014. “The Returns to College Admission for Academically Marginal Students.” *Journal of Labor Economics* 32 (4): 711–54. <https://doi.org/10.1086/676661>.
- Zong, Chen, and Alan Davis. 2024. “Modeling University Retention and Graduation Rates Using IPEDS.” *Journal of College Student Retention: Research, Theory & Practice* 26 (2): 311–33. <https://doi.org/10.1177/15210251221074379>.

A. Local Rental Market Measurement

A natural question is whether on-campus housing is financially attractive relative to the local private rental market. If campus residence is cheaper than the outside option, expansions may improve outcomes partly by lowering students’ housing costs; if it is more expensive, the benefits are more likely to reflect the non-price value of living on campus. Local rental market conditions are measured using the American Community Survey (ACS) 2019 five-year estimates at the Public Use Microdata Area (PUMA) level. PUMAs are non-overlapping statistical geographies constructed to contain at least 100,000 residents, published by the Census Bureau to support microdata dissemination. Three rent variables are used: median gross rent for all unit sizes, median gross rent for one-bedroom units, and median gross rent for two-bedroom units. All are median gross rent (the monthly cash rent plus any utilities paid by the tenant) for occupied units paying cash rent. The per-person cost of sharing a two-bedroom unit is the primary benchmark used throughout, as it most closely approximates the realistic off-campus alternative for most students.

IPEDS reports institutional ZIP codes, not PUMAs. Each institution is matched to a PUMA via a two-step procedure. First, ZIP Code Tabulation Area (ZCTA) polygon boundaries for 2019 are obtained from the Census TIGER/Line files ([U.S. Census Bureau 2019](#)) via the R package “tigris”. The centroid of each ZCTA is then assigned to the PUMA whose polygon contains it. For each institution, the modal five-digit ZIP code observed over the sample period is used as the institutional location, and the corresponding PUMA assignment is looked up from the ZCTA-PUMA crosswalk. This procedure matches 86.9% of institutions to a PUMA; institutions whose ZIP centroids fall in water or near state boundaries may not match and are excluded from rent-comparison analyses.

Table [A.1](#) compares two measures of student housing costs: (i) IPEDS on-campus room charges (converted to monthly equivalent by dividing by nine), and (ii) ACS PUMA-based median gross rent across different unit configurations. Over 2015–2019, the median on-campus monthly charge is approximately \$678. Against the per-person cost of sharing a two-bedroom unit (\$454/month), on-campus is more expensive at roughly 85% of institutions.

On this benchmark, campus residence does not appear to be a cheaper housing option for most students. The fact that students nevertheless de-

mand on-campus housing suggests that it provides non-price benefits, such as proximity to campus, access to peers and services, reduced search costs, or a more structured residential environment.

Table A.1: On-campus room charges compared to ACS local rents, by institution type (medians, 2015–2019).

Institution type	N	On-campus (\$/mo)	ACS all-unit (\$/mo)	ACS 1-bed (\$/mo)	ACS 2-bed (per person, \$/mo)	Ratio: on-campus / 2-bed per person	% on-campus < 2-bed per person
Private	570	683	954	778	484	1.35	17%
Public	361	658	862	663	418	1.42	12%
All	931	678	912	731	454	1.38	15%

Notes:

On-campus monthly charge = annual room charge divided by nine (nine-month academic year). ACS 2-bed pp = per-person cost of sharing a two-bedroom unit. Ratio = on-campus monthly / ACS 2-bed pp; values > 1 indicate on-campus is more expensive. ACS rents are 2019 five-year estimates at the PUMA level.

B. Compositional Stability around Expansion Events

The main graduation regression controls for cohort selectivity directly, and the already-enrolled analysis (Section 9.4) provides indirect evidence that graduation gains are not purely compositional. As a more direct check, I estimate unconditional event studies for the same 46 treatment events used in the difference-in-differences analysis, using institution and cohort fixed effects only, for ten student composition proxies in the stacked panel: Pell grant recipient share (first-time full-time students), a selectivity index, share female, shares Black, Hispanic, White and international, share of degree-seeking undergraduates aged 25 or older, share of first-time students who graduated high school in the past 12 months, and in-state share. The analysis asks whether treated institutions were on a different composition trajectory from controls before the expansion, and whether composition shifted at treatment.

Figure B.1 shows results for the six years prior to expansion and the expansion year and the two years following. Pre-expansion coefficients are small and statistically indistinguishable from zero for nine of the ten outcomes, supporting the parallel trends assumption for student composition. The exception is the share of White students, which shows a significant positive pre-trend at $t - 2$, consistent with a small pre-existing upward drift at treated institutions. This trend reverses after expansion, suggesting it does not reflect a persistent difference in institutional trajectory.

At the expansion year ($t = 0$) - the cohort directly relevant to the main analysis - two outcomes show statistically significant effects. The share of Hispanic students rises by approximately 0.7 percentage points, and the share of students aged 25 or older falls by approximately 0.5 percentage points. Both effects are small in magnitude and run counter to a favorable-selection explanation of the graduation result: if anything, the post-expansion cohort contains a somewhat higher share of students from groups with below-average baseline graduation rates, making it harder to attribute graduation gains to compositional change. The decline in the share of older students is also consistent with the proposed mechanism; additional on-campus housing primarily attracts traditional-age residential students. All other composition measures are statistically flat at $t = 0$.



Figure B.1: Event studies of student composition proxies around housing expansion events. Outcomes include Pell grant share, selectivity, gender, race/ethnicity, age, recent high-school graduation status, and in-state share, all measured for the incoming (first-time) cohort where available. Coefficients are difference-in-differences estimates at each relative year, with year -1 as the omitted reference. Error bars are 95% confidence intervals.

C. Continuous Differencing Models - All Specifications

Table C.1: Effect of Housing Capacity on Retention - First Differences (Per +100 Beds)

	(1)	(2)	(3)	(4)	(5)
Capacity Effect					
Per +100 beds (1-year difference)	-0.007 (0.006)	-0.008 (0.006)	0.002 (0.006)	0.001 (0.006)	-0.003 (0.006)
Controls					
Spending controls		X		X	X
Selectivity index			X	X	X
Cohort size			X	X	
Total enrollment					X
Sample size	15,439	15,439	15,439	15,439	15,439

Notes:

Each column reports first-difference regressions using one-year changes in housing capacity. The coefficient is the percentage-point change in first-year retention per +100 beds. Robust standard errors clustered by institution; *, **, *** indicate significance at 10%, 5%, and 1%.

Table C.2: Effect of Housing Capacity on Graduation - 4-Year Differences (Per +100 Beds Average Exposure)

	(1)	(2)	(3)	(4)	(5)
Capacity Effect					
Per +100 average beds (4-year exposure)	0.097*** (0.024)	0.096*** (0.026)	0.112*** (0.029)	0.112*** (0.029)	0.093*** (0.026)
Controls					
Spending controls		X		X	X
Selectivity index			X	X	X
Cohort size			X	X	
Total enrollment					X
Sample size	9,140	9,140	9,140	9,140	9,140

Notes:

Each column reports first-difference regressions with 4-year cumulative housing exposure. The coefficient is the percentage-point change in 4-year graduation per +100 average beds. Robust standard errors clustered by institution; *, **, *** indicate significance at 10%, 5%, and 1%.

D. Binned Differencing Models - Full Results

Table D.1: Binned Differencing Regressions - Retention

	(1)	(2)	(3)	(4)	(5)
Capacity Effect					
Decrease > 750 beds	-0.036 (0.314)	-0.089 (0.317)	-0.050 (0.313)	-0.136 (0.315)	-0.087 (0.318)
Decrease 750–500 beds	0.153 (0.526)	0.122 (0.526)	0.085 (0.531)	0.036 (0.531)	0.035 (0.534)
Decrease 500–250 beds	0.223 (0.198)	0.206 (0.198)	0.172 (0.196)	0.143 (0.197)	0.176 (0.198)
Increase 250–500 beds	-0.200* (0.106)	-0.223** (0.106)	-0.050 (0.110)	-0.081 (0.110)	-0.141 (0.115)
Increase 500–750 beds	-0.271 (0.189)	-0.295 (0.189)	-0.162 (0.189)	-0.200 (0.190)	-0.201 (0.191)
Increase > 750 beds	-0.131 (0.133)	-0.174 (0.136)	0.126 (0.141)	0.064 (0.142)	-0.032 (0.152)
Controls					
Spending controls		X		X	X
Selectivity index			X	X	X
Cohort size			X	X	
Total enrollment					X
Sample size					
	15,439	15,439	15,439	15,439	15,439

Notes:

First-difference regressions using 1-year changes in housing capacity. Omitted: $|\Delta B| \leq 250$ beds.

Coefficients are percentage-point changes in first-year retention. Robust SEs clustered by institution; *, **, *** denote significance at 10%, 5%, and 1%.

Table D.2: Binned Differencing Regressions - Graduation

	(1)	(2)	(3)	(4)	(5)
Capacity Effect					
Decrease > 750 beds	-1.220 (0.834)	-1.039 (0.862)	-1.520* (0.775)	-1.315 (0.804)	-1.086 (0.829)
Decrease 750–500 beds	-0.033 (0.684)	0.067 (0.688)	-0.258 (0.672)	-0.130 (0.677)	-0.008 (0.671)
Decrease 500–250 beds	-0.056 (0.497)	-0.033 (0.497)	-0.297 (0.503)	-0.256 (0.506)	-0.163 (0.505)
Increase 250–500 beds	0.678*** (0.246)	0.731*** (0.246)	0.808*** (0.244)	0.851*** (0.242)	0.728*** (0.242)
Increase 500–750 beds	1.238*** (0.362)	1.332*** (0.362)	1.480*** (0.364)	1.568*** (0.362)	1.394*** (0.361)
Increase > 750 beds	1.851*** (0.310)	1.928*** (0.343)	2.196*** (0.316)	2.260*** (0.338)	1.942*** (0.335)
Controls					
Spending controls		X		X	X
Selectivity index			X	X	X
Cohort size			X	X	
Total enrollment					X
Sample size					
	9,140	9,140	9,140	9,140	9,140

Notes:

First-difference regressions with 4-year cumulative exposure. Omitted category: $|\Delta B| \leq 250$ beds.

Coefficients are percentage-point changes in 4-year graduation. Robust SEs clustered by institution; *, **, *** denote significance at 10%, 5%, and 1%.

E. Subgroup Analysis

This section describes the subgroup analysis used to explore potential mechanisms. The subgroup splits are intended to capture differences in students' exposure to local housing markets, the relative cost of living on and off campus, and institutional characteristics that may shape how housing expansions affect student outcomes.

If housing expansions operate primarily through a rent market channel, effects should be strongest in settings where students depend more heavily on private rental markets and where off-campus housing costs are particularly salient. This motivates subgroup splits based on local ACS PUMA-level rents, college-reported off-campus living costs, and the socioeconomic composition of the student body. A similar logic applies to urban institutions, where students may be more embedded in local rental markets, and to institutions with different in-state enrollment shares, since these may differ in the extent to which students can commute from the parental home rather than rent locally.

The analysis also considers institutional price and type. Colleges are split by room-and-board charges and tuition levels to assess whether the effects of capacity expansions differ across higher- and lower-cost institutions. I also distinguish between public and private colleges, as well as between institutions that ever require first-year students to live on campus and those that never do so. The latter split addresses the concern that retention or completion effects may differ in settings where on-campus residence is partly determined by institutional policy.

The subgroup indicators are defined at the institution level using means over the sample period. Colleges are classified as low-SES if their average share of Pell Grant recipients is above the sample median, and high-SES otherwise. Colleges are classified as being in high-rent markets if their local median per-person rent (ACS 2019 five-year, per-person cost of sharing a two-bedroom unit at the PUMA level; see Appendix A) exceeds the sample median. Colleges are similarly classified as having high or low college-reported off-campus housing costs (the institution's published cost-of-attendance estimate for off-campus room and board) or high or low on-campus room-and-board charges, according to whether the relevant institution-level mean lies above or below the sample median. The same median split is applied to tuition and in-state enrollment shares. Colleges are classified as urban if they

are located in a city or suburb, and rural otherwise.²¹ Public-private status is based on the institution’s modal control category over the sample period. Finally, the “Not Mandatory” subgroup consists of institutions that never require first-year students to live on campus in any year of the sample period.

The large-small split is reported separately in this appendix, as large and small colleges exhibit very different distributions of absolute capacity changes. This heterogeneity is estimated using proportional capacity-change bins rather than absolute bins, producing more balanced treatment groups. Results are reported in Table E.1.

Table E.1: Effect of Proportional Changes in Capacity on Graduation by Institution Size

	(1) Large (5–10%)	(2) Small (5–10%)
Capacity Effect		
Large increase in 4-year cumulative exposure	0.663*** (0.249)	0.549 (0.367)
Moderate increase in 4-year cumulative exposure	1.044*** (0.276)	-0.242 (0.515)
Moderate decrease in 4-year cumulative exposure	0.017 (0.414)	0.347 (0.918)
Large decrease in 4-year cumulative exposure	-0.702 (0.463)	-0.611 (0.842)
Controls		
Spending controls	X	X
Selectivity index	X	X
Cohort size	X	
Total enrollment		X
Sample size		
	4,880	4,162

Notes: Each column uses first-difference regressions with 4-year cumulative capacity exposure. Capacity changes are binned; the omitted category is an absolute change of less than 5%. Coefficients represent percentage-point changes in graduation relative to this reference. Robust standard errors clustered by institution. *, **, *** denote 10%, 5%, and 1%.

²¹Colleges are defined as “urban” if, based on their address, they are located in a city (large, midsize or small) or a suburb (large, midsize or small) under a methodology developed by the U.S. Census Bureau’s Population Division in 2005. The remaining colleges are defined as rural, and are located in a town (fringe, distant or remote) or a rural area (fringe, distant or remote).

F. Private Housing Supply

If the graduation effects of on-campus housing expansions operate mainly through reduced pressure in the off-campus rental market, then independent increases in local private rental supply should also be associated with higher graduation rates. I test this by adding a measure of local construction to the main binned graduation specification. The measure comes from the Census Bureau’s Building Permits Survey (BPS), which reports the annual dollar value of permits for residential buildings with five or more units at the Core-Based Statistical Area (CBSA) level. CBSAs approximate local housing markets, while five-or-more-unit permits capture the type of large apartment construction most relevant to student rental markets.²²

I match institutions to CBSAs using county Federal Information Processing Standards (FIPS) codes from the Census county-to-CBSA delineation file, obtaining matches for 1,188 of 1,256 institutions. Because there is typically a lag between a permit being issued and the corresponding units becoming available, I use the mean annual permit valuation in the four years before enrollment ($t - 4$ to $t - 1$) as the supply measure. Since permit valuation is already a flow variable (annual permits issued, not a stock of housing), no further differencing is applied; the variable is expressed in hundreds of millions of dollars. The sample is restricted to institution-years with CBSA-level BPS coverage and cohort years for which the full eight-year permit window is observed, leaving cohorts from 2007-2016.

Table F.1 presents three specifications. Column 1 replicates the main binned graduation regression on the BPS-matched subsample; the capacity bin estimates are broadly similar to the full sample, confirming that the subsample is not unusual. Columns 2 and 3 add the lagged private supply measure. The bin estimates in Column 2 are essentially unchanged relative to Column 1. The lagged supply coefficient is small and imprecisely estimated in both columns. The analysis therefore provides no evidence that the main effects are driven by broad CBSA-level multi-family construction trends.

²²I use BPS Table 3av, “Annual Permits by Metropolitan Area: Valuation,” for 2003-2019. The measure is a flow of permitted construction, not a housing-stock measure. It includes all five-or-more-unit residential construction within the CBSA, and may include new student housing or dormitory projects. It should therefore be interpreted as a proxy for broader local rental-supply growth, not as a clean measure of private student rental supply.

Table F.1: Private Housing Supply and Graduation Rate

	Bins only (BPS sample)	Bins + lagged supply	Lagged supply only
Capacity Effect			
$\Delta\text{Beds} < -750$	-1.094 (0.907)	-1.091 (0.903)	
$-750 \leq \Delta\text{Beds} < -500$	0.759 (0.890)	0.758 (0.890)	
$-500 \leq \Delta\text{Beds} < -250$	-0.070 (0.627)	-0.067 (0.625)	
$250 < \Delta\text{Beds} \leq 500$	0.889*** (0.273)	0.888*** (0.273)	
$500 < \Delta\text{Beds} \leq 750$	1.523*** (0.420)	1.518*** (0.420)	
$\Delta\text{Beds} > 750$	2.339*** (0.379)	2.339*** (0.379)	
Priv. supply: lagged avg (\$100m)		-0.009 (0.019)	-0.010 (0.020)
Controls			
Spending controls	X	X	X
Selectivity index	X	X	X
Cohort size	X	X	X
Sample size			
	6,684	6,684	6,684

Notes:

All columns use the graduation binned differencing specification. Column 1 replicates the main binned specification on the BPS-matched subsample. Columns 2 and 3 add the lagged private supply measure: mean annual 5+ unit permit valuation (\$100m) over years t-4 to t-1, representing supply likely available during enrollment given typical permit-to-occupancy lags. Private supply matched to each institution's CBSA via the Census county delineation. Sample restricted to CBSAs with BPS coverage; cohort years 2007–2016. Robust SEs clustered by institution. *, **, *** denote 10%, 5%, 1%.

G. Alternative Bins for Difference Models

Table G.1: Effects of Housing Capacity on Retention and Graduation (Large Bins)

	Ret (1)	Ret (2)	Grad (1)	Grad (2)
Capacity Effect				
Large increase in capacity (>800)	0.066 (0.147)	-0.041 (0.155)	2.237*** (0.343)	1.922*** (0.341)
Medium increase in capacity (400–800)	-0.130 (0.129)	-0.131 (0.137)	1.297*** (0.278)	1.127*** (0.278)
Medium decrease in capacity (-800 to -400)	0.557** (0.282)	0.587** (0.285)	-0.646 (0.517)	-0.544 (0.514)
Large decrease in capacity (<-800)	-0.280 (0.312)	-0.218 (0.313)	-1.132 (0.862)	-0.905 (0.889)
Controls				
Spending controls	X	X	X	X
Selectivity index	X	X	X	X
Cohort size	X		X	
Total enrollment		X		X
Sample size				
	15,439	15,439	9,140	9,140

Notes:

Retention uses 1-year first differences; graduation uses 4-year cumulative housing exposure.

Coefficients are percentage-point changes. Robust SEs clustered by institution;

*, **, *** denote significance at 10%, 5%, and 1%.

H. Proportional Capacity Changes

The proportional differencing operators are defined as

$$\Delta_1^p B_{ic} = \frac{B_{ic} - B_{i,c-1}}{B_{i,c-1}} \quad (\text{H.1})$$

in the retention models, and

$$\Delta_4^p \bar{B}_{ic} = \frac{\bar{B}_{ic} - \bar{B}_{i,c-4}}{\bar{B}_{i,c-4}} \quad (\text{H.2})$$

in the graduation models.

I use two alternative symmetric binning schemes. For retention:

1. 5-10% thresholds: reference group if $|\Delta_1^p B_{ic}| < 0.05$, moderate increase (decrease) if $0.05 \leq \Delta_1^p B_{ic} \leq 0.10$ ($-0.10 \leq \Delta_1^p B_{ic} \leq -0.05$), large increase (decrease) if $\Delta_1^p B_{ic} > 0.10$ ($\Delta_1^p B_{ic} < -0.10$).
2. 10-20% thresholds: reference group if $|\Delta_1^p B_{ic}| < 0.10$, moderate increase (decrease) if $0.10 \leq \Delta_1^p B_{ic} \leq 0.20$ ($-0.20 \leq \Delta_1^p B_{ic} \leq -0.10$), large increase (decrease) if $\Delta_1^p B_{ic} > 0.20$ ($\Delta_1^p B_{ic} < -0.20$).

The same binning system is used for graduation models, but applied to $\Delta_4^p \bar{B}_{ic}$. The use of proportional changes does introduce the risk of results being skewed by spurious “large” percentage changes due to low capacity in $c - 1$. Two safeguards are used to reduce this risk. 1) Observations where $B_{c-1} < 50$ (or, in the case of graduation models, $\bar{B}_{i,c-4} < 50$) are excluded from the analysis. 2) Observations with $|B_{ic} - B_{i,c-1}| \leq 100$ (or, in the case of graduation models, $|\bar{B}_{ic} - \bar{B}_{i,c-4}| \leq 100$) are placed in the reference group regardless of $\Delta_1^p B_{ic}$ (or $\Delta_4^p \bar{B}_{ic}$).

The estimates for the effect on retention, as in the main analysis, are statistically indistinguishable from zero. For graduation, proportional increases in capacity deliver positive, statistically significant gains under the 5–10% scheme: point estimates are roughly 0.6–1.1 percentage points. Proportional decreases are statistically indistinguishable from zero. Under the 10–20% scheme (which has a larger reference group), only moderate increases remain significant, with an effect of roughly 0.5 percentage points. All other bins are imprecisely estimated (the effect of large expansions is positive, as expected and consistent with the results throughout other specifications, but imprecisely estimated). The proportional analysis supports the main finding that capacity expansions raise on-time completion, though magnitudes attenuate

when the reference window widens. The smaller magnitudes are consistent with two explanations. First, proportional measures remain more sensitive to variation in denominators. Second, Safeguard 2 mechanically attenuates the estimates by placing some genuinely large proportional expansions (those with absolute changes ≤ 100 beds) into the reference category.

Table H.1: Effect of Proportional Changes in Capacity on Retention .

	(1) 5–10%	(2) 5–10%	(3) 10–20%	(4) 10–20%
Capacity Effect				
Large increase in capacity	-0.063 (0.104)	-0.099 (0.105)	-0.053 (0.165)	-0.088 (0.166)
Moderate increase in capacity	0.109 (0.097)	0.081 (0.099)	-0.090 (0.128)	-0.124 (0.129)
Moderate decrease in capacity	0.166 (0.180)	0.175 (0.179)	-0.092 (0.242)	-0.094 (0.246)
Large decrease in capacity	0.107 (0.212)	0.117 (0.214)	0.444 (0.421)	0.479 (0.423)
Controls				
Spending controls	X	X	X	X
Selectivity index	X	X	X	X
Cohort size	X		X	
Total enrollment		X		X
Sample size				
	15,208	15,208	15,208	15,208

Notes: Each column reports coefficients from first-difference regressions. Reference category is small or no change in housing capacity (absolute change of less than 5 percent in columns 1 and 2, and absolute change of less than 10 percent in columns 3 and 4). Robust standard errors clustered by institution in parentheses. *, **, *** indicate significance at 10%, 5%, and 1%, respectively.

Table H.2: Effect of Proportional Changes in Capacity on Graduation

	(1) 5–10%	(2) 5–10%	(3) 10–20%	(4) 10–20%
Capacity Effect				
Large increase in 4-year cumulative exposure	0.625*** (0.211)	0.510** (0.210)	0.321 (0.275)	0.199 (0.276)
Moderate increase in 4-year cumulative exposure	1.079*** (0.249)	1.014*** (0.248)	0.541** (0.226)	0.457** (0.225)
Moderate decrease in 4-year cumulative exposure	0.606 (0.388)	0.685* (0.392)	-0.172 (0.555)	-0.077 (0.553)
Large decrease in 4-year cumulative exposure	-0.207 (0.447)	-0.125 (0.446)	-0.767 (0.675)	-0.709 (0.677)
Controls				
Spending controls	X	X	X	X
Selectivity index	X	X	X	X
Cohort size	X		X	
Total enrollment		X		X
Sample size				
	9,042	9,042	9,042	9,042

Notes: Each column reports coefficients from first-difference regressions. Reference category is small or no change in housing capacity (absolute change of less than 5 percent in columns 1 and 2, and absolute change of less than 10 percent in columns 3 and 4). Robust standard errors clustered by institution in parentheses. *, **, *** indicate significance at 10%, 5%, and 1%, respectively.

I. Start-to-start Difference Models

Table I.1: Binned Differencing - Graduation (Start-to-Start)

	Spec 4	Spec 5
Capacity Effect		
Decrease > 750 beds	-0.123 (0.573)	-0.083 (0.570)
Decrease 750–500 beds	-0.084 (0.552)	0.001 (0.552)
Decrease 500–250 beds	0.878** (0.424)	0.916** (0.423)
Increase 250–500 beds	0.638*** (0.245)	0.565** (0.244)
Increase 500–750 beds	1.324*** (0.295)	1.172*** (0.294)
Increase > 750 beds	1.913*** (0.341)	1.672*** (0.346)
Controls		
Spending controls	X	X
Selectivity index	X	X
Cohort size	X	
Total enrollment		X
Sample size		
	9,145	9,145

Notes:

First-difference regressions using 4-year start-to-start changes in housing capacity.

Omitted: $|\Delta B| \leq 250$ beds. Coefficients are percentage-point changes in 4-year graduation.

Robust SEs clustered by institution; *, **, *** indicate significance at 10%, 5%, and 1%.

J. Graduate Count Model

Table J.1: Effect of Housing Capacity on Graduate Count — 4-Year Differences (Per +1 Average Bed)

	(1)	(2)	(3)	(4)	(5)
Capacity Effect					
Per +1 average bed (4-year exposure)	0.160*** (0.024)	0.113*** (0.023)	0.077*** (0.014)	0.051*** (0.014)	0.092*** (0.021)
Controls					
Spending controls		X		X	X
Selectivity index			X	X	X
Cohort size			X	X	
Total enrollment					X
Sample size					
	9,140	9,140	9,140	9,140	9,140

Notes:

Outcome is the 4-year change in the number of graduates. Coefficients should be interpreted as additional graduates per +1 average bed over the cohort window. Cohort size and total enrollment enter as controls (not denominators) so that gains are not driven mechanically by enrollment increases. Robust SEs clustered by institution; *, **, *** indicate 10%, 5%, and 1%.

K. Difference-in-Differences: Alternative Treatment Thresholds

Table K.1: Robustness of matched graduation DiD to treatment detection thresholds

	Graduation (matched K=10)				
	(1) Most restrictive	(2) Slightly restrictive	(3) Main	(4) Slightly permissive	(5) Most permissive
Post x Treated	0.605 (1.333)	1.508* (0.778)	1.416** (0.604)	1.571** (0.631)	1.528*** (0.508)
Controls:					
Spending controls	X	X	X	X	X
Selectivity index	X	X	X	X	X
Cohort size	X	X	X	X	X
Year fixed effects	X	X	X	X	X
College fixed effects	X	X	X	X	X
Observations	215	573	757	939	1302
Treatment events	9	31	46	62	117
Specification:					
Change in beds (abs)	500	400	300	250	200
Change in beds (percent)	0.1	0.1	0.1	0.08	0.05
Change in beds (student ratio)	0.05	0.025	0.025	0.02	0

Notes:

Each column reports coefficients from difference-in-differences regressions under alternative treatment definitions. Robust standard errors clustered by institution in parentheses. *, **, *** indicate significance at 10%, 5%, and 1%, respectively.

L. Individual Treatment Events

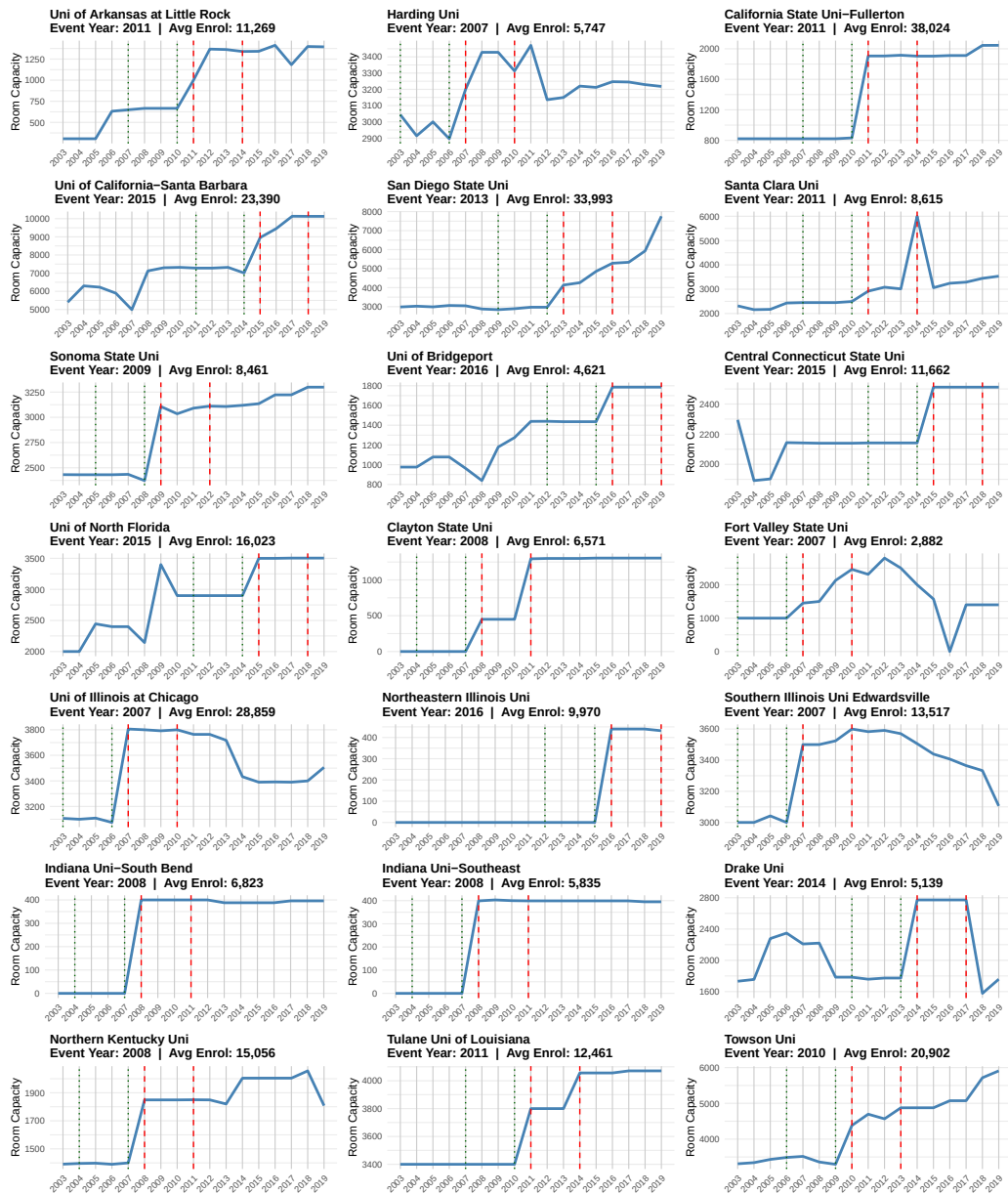
Table L.1 lists the individual treatment events used in the difference-in-differences analysis. “Capacity t ” refers to capacity in the treatment year, while $t - 4$ and $t - 1$ represent the pre-treatment capacity for graduation and retention outcomes respectively. Capacity in $t + 3$ is shown where available to verify that each expansion was sustained rather than temporary. The figures that follow the table plot the expansion paths in each treated college; the red dashed lines represent year t (the year of treatment) and year $t + 3$ (the year in which the cohort enrolling in year t graduated), while the green dashed lines represent year $t - 1$ (the pre-treatment year for retention models) and year $t - 4$ (the pre-treatment year for graduation models). This yields 51 treatment events that fit the criteria, fall within the sample period, and have available matched controls. Of these, 46 are used in the graduation models.

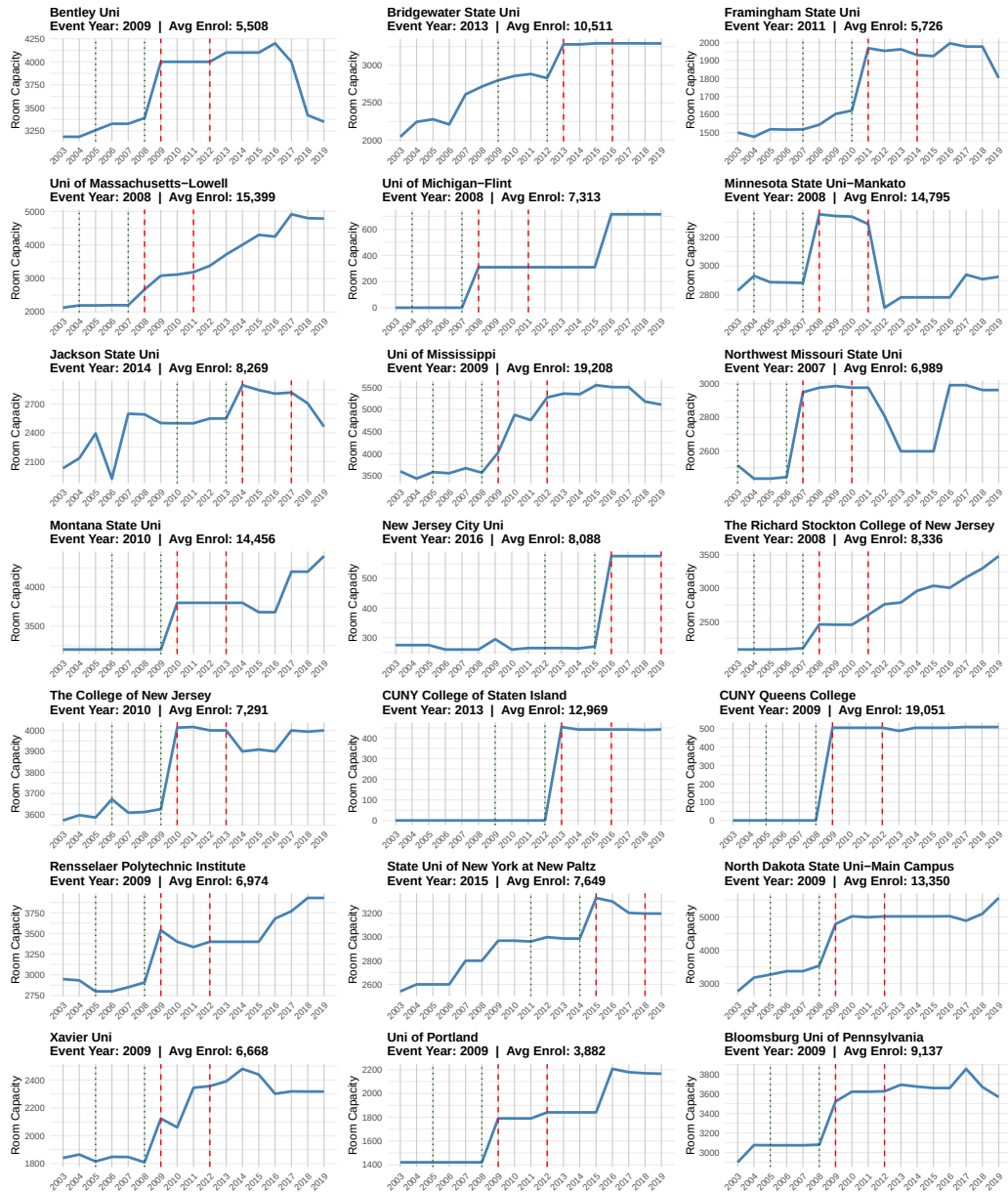
Table L.1: Capacity around treatment events (treated colleges used in the analysis).

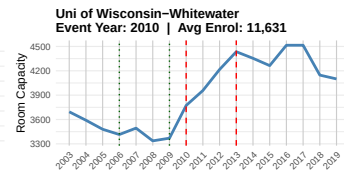
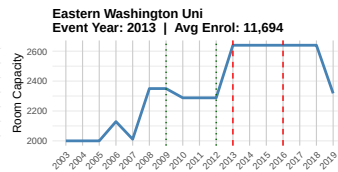
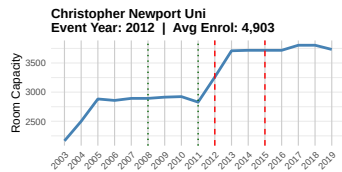
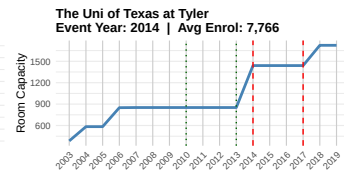
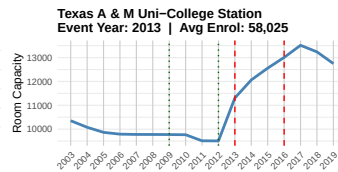
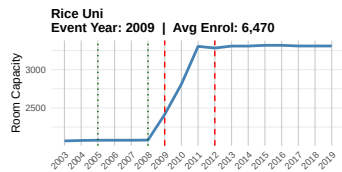
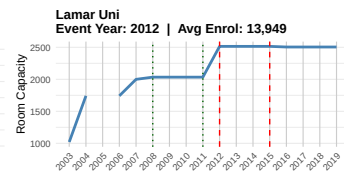
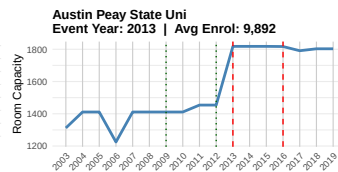
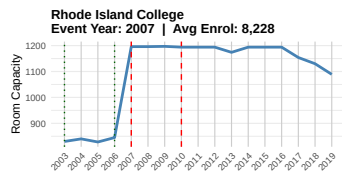
Institution	Event year	Capacity t-4	Capacity t-1	Capacity t	Capacity t+3
Austin Peay State Uni	2013	1411	1454	1818	1817
Bentley College	2009	3258	3390	4000	4000
Bloomsburg Uni of Pennsylvania	2009	3074	3080	3523	3627
Bridgewater State College	2013	2801	2829	3281	3295
CLAYTON COLLEGE AND STATE Uni	2008	0	0	450	1295
CUNY Queens College	2009	0	0	506	506
California State Uni-Fullerton	2011	821	832	1906	1904
Central Connecticut State Uni	2015	2142	2142	2512	2512
Christopher Newport Uni	2012	2893	2829	3259	3718
College of Staten Island CUNY	2013	0	0	454	442
Drake Uni	2014	1784	1773	2772	2772
Eastern Washington Uni	2013	2350	2288	2640	2640
Fort Valley State Uni	2007	1000	1000	1450	2463
Framingham State College	2011	1517	1622	1969	1931
Harding Uni	2007	3044	2896	3200	3313
Indiana Uni-South Bend	2008	0	0	400	400
Indiana Uni-Southeast	2008	0	0	399	399
Jackson State Uni	2014	2500	2550	2900	2823
Lamar Uni	2012	2034	2034	2515	2515
Minnesota State Uni-Mankato	2008	2932	2883	3355	3288
Montana State Uni	2010	3200	3200	3800	3800
New Jersey City Uni	2016	265	270	576	576
North Dakota State Uni-Main Campus	2009	3268	3535	4784	5011
Northeastern Illinois Uni	2016	0	0	440	432
Northern Kentucky Uni	2008	1396	1400	1849	1850
Northwest Missouri State Uni	2007	2514	2444	2950	2977

Table L.1: Capacity around treatment events (treated colleges used in the analysis). (*continued*)

Institution	Event year	Capacity t-4	Capacity t-1	Capacity t	Capacity t+3
Rensselaer Polytechnic Institute	2009	2800	2908	3540	3400
Rhode Island College	2007	830	845	1196	1194
Rice Uni	2009	2078	2080	2415	3282
San Diego State Uni	2013	2841	2969	4135	5285
Santa Clara Uni	2011	2450	2494	2919	6002
Sonoma State Uni	2009	2431	2373	3107	3111
Southern Illinois Uni-Edwardsville	2007	3000	3000	3500	3600
State Uni of New York at New Paltz	2015	2962	2987	3330	3198
Stockton Uni	2008	2080	2100	2458	2600
Texas A and M Uni	2013	9772	9505	11308	13008
The College of New Jersey	2010	3673	3626	4013	4000
The Uni of Texas at Tyler	2014	852	852	1440	1440
Towson Uni	2010	3487	3296	4381	4875
Tulane Uni of Louisiana	2011	3400	3400	3800	4056
Uni of Arkansas at Little Rock	2011	649	666	1000	1340
Uni of Bridgeport	2016	1440	1436	1786	1786
Uni of California-Santa Barbara	2015	7279	7010	8959	10129
Uni of Illinois at Chicago	2007	3107	3075	3806	3800
Uni of Massachusetts-Lowell	2008	2193	2199	2670	3188
Uni of Michigan-Flint	2008	0	0	310	310
Uni of Mississippi	2009	3586	3573	4031	5268
Uni of North Florida	2015	2900	2900	3500	3504
Uni of Portland	2009	1421	1421	1790	1841
Uni of Wisconsin-Whitewater	2010	3415	3370	3773	4435
Xavier Uni	2009	1815	1809	2123	2356







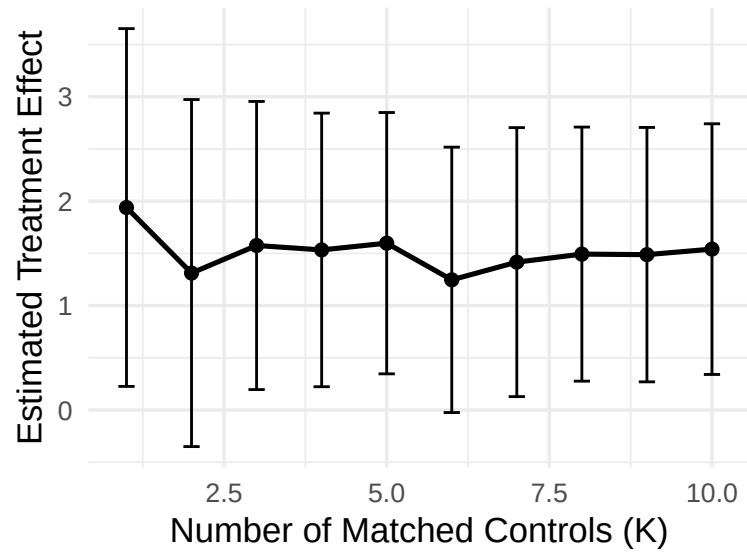


Figure M.1: Estimated treatment effect of capacity expansions on graduation rates for different values of K (number of matched controls per treatment event)

M. Sensitivity of Matched Estimates to Number of Matched Controls

N. Leave-One-Out Difference-in-Differences

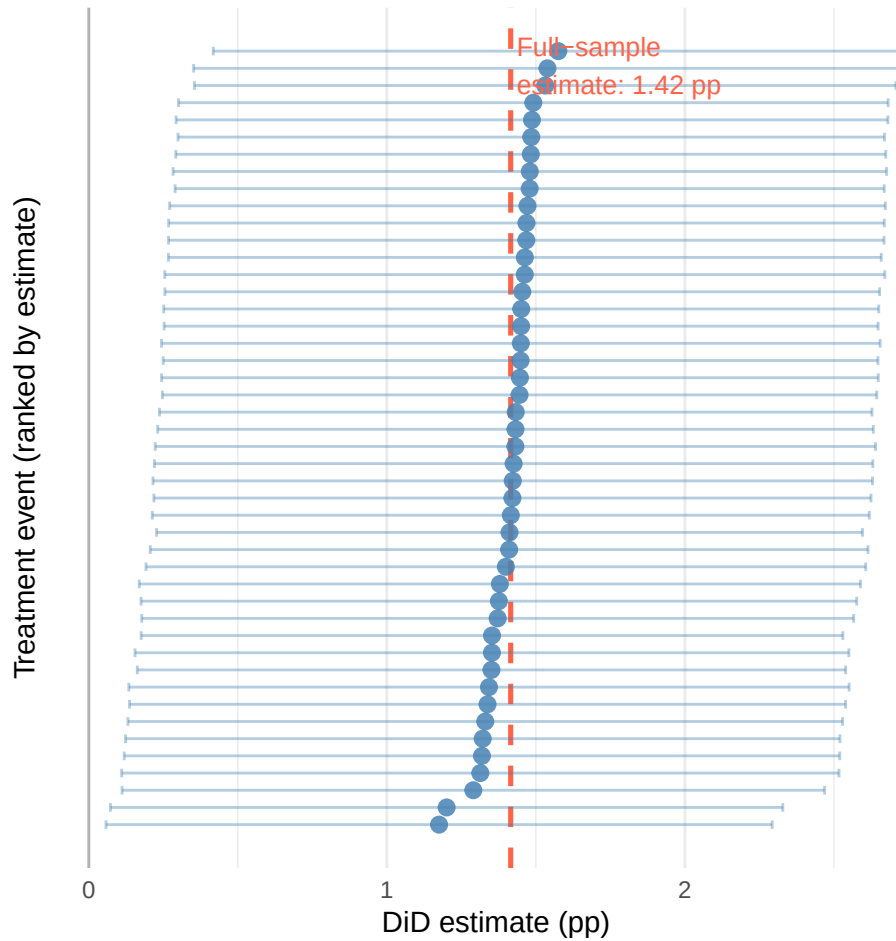


Figure N.1: Leave-one-out sensitivity of the matched difference-in-differences graduation estimate. Each point re-estimates the main specification after removing one treatment event; the vertical line shows the full-sample estimate. Bars are 95% confidence intervals.

Figure N.1 reports leave-one-out graduation difference-in-differences estimates. Each point is the treatment effect re-estimated after dropping one of the 46 treatment events. Estimates are sorted by value; the vertical dashed line marks the full-sample estimate. Point estimates range from 1.18 to 1.58 pp and remain significant at the 5% level in all 46 cases, confirming that no single institution drives the result.